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Observer-tracked privacy display

Abstract

A display device comprising a spatial light modulator having a display polariser arranged on one side is provided with an additional polariser arranged on the same side as the display polariser and a polar control retarder between the additional polariser and the display polariser. The polar control retarder includes a liquid crystal retarder having two surface alignment layers disposed adjacent to a layer of liquid crystal material on opposite sides. The surface alignment layers provide alignment in the adjacent liquid crystal material with a twist. The out-of-plane orientation of the twisted layer of liquid crystal material is modified across at least one region of the display device to provide a transmission function in response to the measured location of an off-axis snooper, achieving increased size of polar region for which desired uniformity of security factor, or reduced distraction across the display to the driver in an automotive application is achieved.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure generally relates to luminance uniformity in display devices, including for application to privacy display.

BACKGROUND

[0002] Display devices may be configured to reduce luminance in directions offset from a viewing axis, typically normal to the display device. Such off-axis luminance reduction may have various purposes. In one example, off-axis luminance reduction may be applied in a privacy display to provide visibility of a displayed image to a primary user in an on-axis position and reduce visibility of the displayed image to a snooper in an off-axis position. In another example off-axis luminance reduction may be applied to reduce stray light that may otherwise cause unwanted reflections in an environment around the display device, for example within an automobile.

[0003] A privacy function may be provided by micro-louvre optical films that transmit a high luminance from a display device in an on-axis direction with low luminance in off-axis positions. However, such films are not switchable, and thus the display is limited to privacy-only function.

[0004] Switchable privacy displays may be provided by implementing control of the off-axis optical output. Control of off-axis privacy may be provided by means of contrast reduction, for example by adjusting the liquid crystal bias tilt in an In-Plane-Switching LCD.

[0005] Control may be further provided by means of off-axis luminance reduction. Luminance reduction may be achieved by means of switchable backlights for a liquid crystal display (LCD) spatial light modulator. Off-axis luminance reduction may also be provided by switchable liquid crystal retarders and compensation retarders arranged to modulate the input and/or output directional luminance profile of a spatial light modulator.

[0006] Control may be further provided by means of off-axis reflectivity increase. Reflectivity increase may be achieved by means of switchable liquid crystal retarders, compensation retarders that are arranged to control the polarisation of ambient light that falls onto a reflective polariser.

[0007] However, in operation such off-axis luminance reduction may provide non-uniformity of luminance across the area of the display device for display viewers. For many applications, it would be desirable to increase the perceived uniformity.

BRIEF SUMMARY

[0008] According to the present disclosure, there is provided a display apparatus comprising: a display device comprising: a spatial light modulator arranged to output spatially modulated light; a display polariser arranged on a side of the spatial light modulator, the display polariser being a linear polariser; and a view angle control arrangement comprising: an additional polariser arranged

on the same side of the spatial light modulator as the display polariser, the additional polariser being a linear polariser; and at least one polar control retarder arranged between the display polariser and the additional polariser, wherein the at least one polar control retarder comprises a switchable liquid crystal retarder comprising a layer of liquid crystal material; two surface alignment layers disposed adjacent to the layer of liquid crystal material and on opposite sides thereof, the surface alignment layers each being arranged to provide homogenous alignment in the adjacent liquid crystal material, the liquid crystal material being provided with a twist; and first and second electrode arrangements disposed on opposite sides of the layer of liquid crystal material and arranged to provide an electric field perpendicular to the layer of liquid crystal material, wherein the display apparatus further comprises a control system arranged to control the spatial light modulator and to supply voltages to the first and second electrode arrangements for providing the electric field perpendicular to the layer of liquid crystal material, the control system being arranged to vary the voltages to the first and second electrode arrangements for controlling a direction of minimum light transmission of the view angle control arrangement.

[0009] The display apparatus may advantageously provide controllable luminance for desirable observer locations. In a privacy mode of operation of the display apparatus, the size of polar region for which high image security is achieved may advantageously be increased. The display apparatus may be switched to a share mode of operation for high image visibility for multiple display users.

[0010] The display apparatus may further comprise a viewer tracking system arranged to track the location of a viewer, wherein the control system may be arranged to vary the voltages to the first and second electrode arrangements in response to the tracked location of the viewer. The illumination provided to a moving observer may be optimised for uniformity of luminance or for uniformity of security factor. Advantageously image visibility to a driver in an automotive vehicle may be minimised.

[0011] The control system may be arranged to accept user input indicating a direction of minimum light transmission of the view angle control arrangement, and the control system may be arranged to vary the voltages to the first and second electrode arrangements in response to the user input. Advantageously the cost of the control system may be reduced.

[0012] Components of the optical axis of the liquid crystal material in the plane of the layer of liquid crystal material may have an average direction that may be along the predetermined direction. Advantageously luminance may be controlled towards an off-axis observer of the display device.

[0013] The magnitude of the electric field perpendicular to the layer of liquid crystal material may change monotonically along a predetermined axis across at least part of the display device. Advantageously, a display apparatus may be provided which reduces changes in luminance across at least part of the display device as perceived by a viewer in a predetermined position, in at least one state of the liquid crystal material. For example, for an off-axis viewer, e.g. a snooper, the display apparatus may provide a more uniform luminance across at least part of the display, e.g. across the width of the display. In such cases, the luminance may be a uniform reduced luminance such that the off-axis viewer cannot observe any image output by the display device at any point along the display device. Further the uniformity of the security level may be increased for a snooper. In a similar manner, for an on-axis viewer, e.g. the primary user of the display device, the display apparatus may also provide a more uniform luminance across at least part of the display for the on-axis viewer. Such uniform luminance across at least part of the display apparatus may be a non-reduced, or observable, luminance such that the on-axis viewer can observe the image output by the display. This may therefore reduce the “roll-off” which occurs on prior art displays whereby the luminance reduces towards the edges of the display for an on-axis viewer. Advantageously display device users may have increased luminance and image uniformity. Transmission at desirable viewing angles from the display is increased.

[0014] The magnitude of the electric field may change monotonically along a predetermined axis

across at least part of the display device so that directions of minimum light transmission of the view angle control arrangement from points of said at least part of the display device may be directed towards a common off-axis point in front of the display device. Advantageously an observer near to the common off-axis point may see low intensity across the width of at least part of the display. The viewing angle of the display changes monotonically along the predetermined direction, and the electric field change advantageously achieves a matched reduction in transmission corresponding to the monotonic change in viewing angle. Uniformity of security factor is increased and image visibility reduced. In an automotive vehicle, driver distraction may be minimised.

[0015] The first electrode arrangement may comprise a continuous electrode extending over the entirety of the layer of liquid crystal material, the continuous electrode having at least one contact disposed at each of opposite ends of the continuous electrode along the predetermined axis and arranged to supply respective voltages that create a voltage profile on the continuous electrode along the predetermined axis for providing the electric field perpendicular to the layer of liquid crystal material. Advantageously the cost and complexity of the electrode may be reduced and desirable transmission of polar control retarder with polar angle may be achieved.

[0016] At least one of the contacts may comprise a conductive bar extending perpendicular to the predetermined axis. Advantageously uniformity of transmission reduction in the predetermined direction may be increased.

[0017] The first electrode arrangement may comprise plural electrodes separated along the predetermined axis, the plural electrodes being arranged to supply different voltages that may change monotonically along the predetermined axis. The location of the common point may be modified.

[0018] The second electrode arrangement may be arranged to supply a common voltage. The second electrode arrangement may comprise a continuous electrode extending over the entirety of the layer of liquid crystal material. Advantageously cost and complexity may be reduced.

[0019] The magnitude of the electric field may be uniform across the entirety of the display device. Each of the first electrode arrangement and the second electrode arrangement may comprise a continuous electrode extending over the entirety of the layer of liquid crystal material. The size of the polar region for high image security and high luminance may be increased. Advantageously the cost and complexity of the electrodes and control system may be reduced.

[0020] The surface alignment layers may each be arranged to provide homogenous alignment in the adjacent liquid crystal material, wherein the surface alignment layers may have an in-plane component in the plane of the layer of liquid crystal material having an angle that changes monotonically along a predetermined axis across at least part of the display device. Further improvement of uniformity across the display device may be achieved. Cost and complexity of electrodes and the control system may be achieved.

[0021] The switchable liquid crystal retarder may have a retardance for light of a wavelength of 550 nm in a range from 300 nm to 1500 nm, preferably in a range from 400 nm to 1200 nm. The twist may be in a range from 60° to 120°, preferably in a range from 70° to 90°. Advantageously in a privacy mode of operation off-axis transmission reduction may be provided directed towards typical off-axis user locations.

[0022] The electric vector transmission direction of the additional polariser may be oriented at a non-zero angle with respect to the electric vector transmission direction of the display polariser. The non-zero angle may be the same as the twist. Advantageously transmission of light towards an off-axis user may be minimised.

[0023] The surface alignment layer closest to the display polariser may have a pretilt having a pretilt direction with a component in the plane of the layer of liquid crystal material that may be parallel or perpendicular to an electric vector transmission direction of the display polarisers, and the surface alignment layer closest to the additional polariser may have a pretilt having a pretilt

direction with a component in the plane of the layer of liquid crystal material that may be parallel or perpendicular to an electric vector transmission direction of the additional polariser.

Advantageously transmission may be minimised over an increased size of polar viewing region.

[0024] The at least one polar control retarder may further comprise at least one passive retarder.

The at least one passive retarder may be an A-plate with a retardance for light of a wavelength of 550 nm in a range of 250 nm to 300 nm. Advantageously a spatial light modulator with display polarisers not aligned to the desirable alignment layer directions may be provided.

[0025] The liquid crystal material may have a positive dielectric anisotropy. Advantageously a share mode operation may be provided at zero volts for low power consumption.

[0026] The display apparatus may be curved with a concave curvature as viewed from the output side. Advantageously increased luminance uniformity and increased uniformity of security factor may be achieved. The display apparatus may be conveniently provided with curved mounting surfaces, including surfaces of an automotive cabin.

[0027] Said display polariser may be an output display polariser arranged on the output side of the spatial light modulator. Advantageously the thickness and complexity of the backlight may be reduced. The polar control retarder may be retrofitted to an assembled display module.

[0028] There may be no reflective polariser arranged between the output display polariser and the at least one polar control retarder. Advantageously cost and complexity may be reduced.

[0029] The display apparatus may further comprise a reflective polariser arranged between the output polariser and the at least one polar control retarder, the reflective polariser being a linear polariser arranged to pass the same linearly polarised polarisation component as the output polariser. Advantageously the security factor of the display may be increased and driver distraction reduced.

[0030] The spatial light modulator may comprise an emissive spatial light modulator arranged to emit the spatially modulated light. Advantageously image contrast may be increased and display thickness reduced.

[0031] The spatial light modulator may comprise a transmissive spatial light modulator and the display apparatus may further comprise a backlight arranged to illuminate the spatial light modulator.

[0032] The spatial light modulator may comprise a transmissive spatial light modulator, the display apparatus may further comprise a backlight arranged to illuminate the spatial light modulator, and said display polariser may be an input display polariser arranged on the input side of the spatial light modulator. Advantageously a backlight with desirable luminance profile for use in privacy applications may be provided. Front reflections to a display user may be reduced and image contrast improved.

[0033] The display apparatus may further comprise: an output display polariser arranged on the output side of the spatial light modulator; a further additional polariser arranged on the output side of the spatial light modulator, the further additional polariser being a linear polariser; and at least one further polar control retarder arranged between the further additional polariser and the output display polariser.

[0034] The luminance of light for an off-axis viewer may advantageously be further reduced thereby further improving the privacy function of the display device. Each of the polar control retarders, together with their associated polarisers, may independently reduce the luminance in an off-axis direction. Together the two polar control retarders and their associated polarisers may have a multiplicative effect in reducing the off-axis luminance. This may therefore significantly reduce the luminance for an off-axis viewer.

[0035] It may be possible to advantageously reduce changes in luminance across at least part of the display device in two directions perpendicular to one another. This may, for example, make it possible to create uniform luminance for an off-axis viewer in two different directions relative to the display device, e.g. for an off-axis viewer to the side of the display device and for an off-axis

viewer at an angle above or below the display device. When used to provide uniform reduced luminance, this may therefore improve the privacy achieved by the display device.

[0036] It may be possible to have an off-axis privacy effect that is controlled between two different predetermined lateral directions. The predetermined directions may be opposite to each other to provide switchable luminance reduction on either side of the display by means of control of which polar control retarder is operated.

[0037] The at least one further polar control retarder may comprise a switchable liquid crystal retarder comprising a layer of liquid crystal material, and the display apparatus may further comprise first and second electrode arrangements disposed on opposite sides of the layer of liquid crystal material, wherein first and second electrode arrangements may be arranged to provide an electric field perpendicular to the layer of liquid crystal material, wherein the magnitude of the electric field may change monotonically along a predetermined axis across at least part of the display device. The electric fields may have different profiles and may increase the size of the polar region for which high image security and low driver distraction is achieved.

[0038] The display apparatus may further comprise: a further additional polariser arranged on the same side of the spatial light modulator as the additional polariser outside the additional polariser, the further additional polariser being a linear polariser; and at least one further polar control retarder arranged between the additional polariser and the further additional polariser. The polar control retarders and additional polarisers may be provided as a single component, advantageously reducing complexity of assembly. The polar control retarders may be provided between the backlight and the spatial light modulator, advantageously reducing the visibility of front reflections to the user and increasing perceived image contrast.

[0039] The at least one further polar control retarder may comprise a switchable liquid crystal retarder comprising a layer of liquid crystal material, and the display apparatus may further comprise first and second electrode arrangements disposed on opposite sides of the layer of liquid crystal material, wherein first and second electrode arrangements may be arranged to provide an electric field perpendicular to the layer of liquid crystal material, wherein the magnitude of the electric field may change monotonically along a predetermined axis across at least part of the display device. Advantageously the uniformity of luminance reduction from the further polar control retarder is increased. Uniformity of security factor may be increased.

[0040] The backlight may comprise: at least one first light source arranged to provide input light; at least one second light source arranged to provide input light in an opposite direction from the at least one first light source; a waveguide arrangement comprising at least one waveguide, the waveguide arrangement being arranged to receive the input light from the at least one first light source and the at least one second light source and to cause light from the at least one first light source and the at least one second light source to exit from the waveguide arrangement by breaking total internal reflection; and an optical turning film component comprising: an input surface arranged to receive the light exiting from a waveguide through a light guiding surface of the waveguide by breaking total internal reflection, the input surface extending across the plane; and an output surface facing the input surface, wherein the input surface may comprise an array of prismatic elements. Luminance profiles may be provided with first and second maxima with different polar directions. The luminance directed to said first and second maxima may be controlled by means of control of the light sources. Advantageously the image security provided to a user such as a driver may be increased by control of at least one of the light sources. High brightness and efficiency may be provided for the primary display user.

[0041] The waveguide arrangement may comprise: a waveguide extending across a plane and comprising: first and second opposed light guiding surfaces arranged to guide light along the optical waveguide, the second light guiding surface being arranged to guide light by total internal reflection, and first and second input ends arranged between the first and second light guiding surfaces and extending in a lateral direction between the first and second light guiding surfaces;

wherein the at least one first light source may be arranged to input light into the waveguide through the first input end and the at least one second light source may be arranged to input light into the waveguide through the second input end, and the waveguide may be arranged to cause light from the at least one first light source and the at least one second light source to exit from the waveguide through one of the first and second light guiding surfaces by breaking total internal reflection. Advantageously the thickness and complexity of the waveguide is reduced.

[0042] The waveguide arrangement may comprise: a first waveguide extending across a plane and comprising first and second opposed light guiding surfaces arranged to guide light along the optical waveguide, the second light guiding surface being arranged to guide light by total internal reflection; and a first input end arranged between the first and second light guiding surfaces and extending in a lateral direction between the first and second light guiding surfaces; wherein the at least one first light source may be arranged to input light into the first waveguide through the first input end, and the first waveguide may be arranged to cause light from the at least one first light source to exit from the first waveguide through one of the first and second light guiding surfaces by breaking total internal reflection; a second waveguide extending across the plane arranged in series with the first waveguide and comprising first and second opposed light guiding surfaces arranged to guide light along the optical waveguide, the second light guiding surface being arranged to guide light by total internal reflection, and a second input end arranged between the first and second light guiding surfaces and extending in a lateral direction between the first and second light guiding surfaces; wherein the at least one second light source may be arranged to input light into the second waveguide through the second input end, and the second waveguide may be arranged to cause light from the at least one second light source to exit from the second waveguide through one of the first and second light guiding surfaces by breaking total internal reflection, and wherein the first and second waveguides may be oriented so that at least one first light source and at least one second light source input light into the first and second waveguides in opposite directions. Advantageously the uniformity of light output from each waveguide can be increased for each of the display users.

[0043] The display apparatus may comprise plural, tiled display devices. Advantageously increased display area may be provided at increased display yield and lower display cost. Replacement of regions across the display device arising from failure or damage to small areas may be conveniently provided with lower cost. Illumination characteristics of different regions of the display device may be provided by different display device structures.

[0044] The magnitude of the electric field across at least part of the display device may be different in different display devices. Some or all of the illumination structures from respective display devices may be modified in response to user location. Advantageously increased size of polar region for high uniformity of luminance and/or security factor may be achieved.

[0045] The profile of the change in the magnitude of the electric field monotonically along a predetermined axis across at least part of the display device may be different in different display devices. The directions of minimum light transmission of the view angle control arrangement from points of said at least part of each of the display devices may be directed towards a common off-axis point in front of the display device. Advantageously luminance uniformity of each of the display devices to respective users may be increased. Uniformity of security factor may be increased over a wider polar region and wider total display width.

[0046] Embodiments of the present disclosure may be used in a variety of optical systems. The embodiments may include or work with a variety of projectors, projection systems, optical components, displays, microdisplays, computer systems, processors, self-contained projector systems, visual and/or audio-visual systems and electrical and/or optical devices. Aspects of the present disclosure may be used with practically any apparatus related to optical and electrical devices, optical systems, presentation systems or any apparatus that may contain any type of optical system. Accordingly, embodiments of the present disclosure may be employed in optical systems, devices used in visual and/or optical presentations, visual peripherals and so on and in a number of

computing environments.

[0047] Before proceeding to the disclosed embodiments in detail, it should be understood that the disclosure is not limited in its application or creation to the details of the particular arrangements shown, because the disclosure is capable of other embodiments. Moreover, aspects of the disclosure may be set forth in different combinations and arrangements to define embodiments unique in their own right. Also, the terminology used herein is for the purpose of description and not of limitation.

[0048] These and other advantages and features of the present disclosure will become apparent to those of ordinary skill in the art upon reading this disclosure in its entirety.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0049] Embodiments are illustrated by way of example in the accompanying FIGURES, in which like reference numbers indicate similar parts, and in which:

[0050] FIG. 1A is a schematic diagram illustrating a schematic top view of a privacy display for use by a passenger in an automotive vehicle;

[0051] FIG. 1B is a schematic diagram illustrating a schematic top view of a privacy display for use in off-axis use applications;

[0052] FIG. 2A is a schematic side perspective view of a display device providing uniformity for a display user;

[0053] FIG. 2B is a schematic side view of an optical stacking for the display device of FIG. 2A;

[0054] FIG. 2C is a front view of the stack of layers of the display device of FIG. 2A;

[0055] FIG. 3A is a perspective view of a polar control retarder that may be applied in FIG. 2A, comprising a twisted nematic switchable liquid crystal (LC) retarder and profile of voltage in the lateral direction across at least one region of the polar control retarder in a privacy mode of operation;

[0056] FIG. 3B is a side view of propagation of output light from a SLM through the optical stack of FIG. 3A;

[0057] FIG. 3C is a perspective view of a polar control retarder that may be applied in FIG. 2A, comprising a twisted nematic switchable LC retarder and uniform voltage in the lateral direction across at least one region of the polar control retarder in a privacy mode of operation;

[0058] FIG. 3D is a schematic perspective side view of a switchable liquid crystal retarder that may be applied in FIG. 2A, comprising a twisted nematic switchable LC retarder with alignment layer orientations that vary across the lateral direction, and further comprising a profile of voltage in the lateral direction across at least one region of the polar control retarder in a privacy mode of operation;

[0059] FIG. 3E is a graph of in-plane angle against position in the predetermined direction and illustrates exemplary profiles of the in-plane components provided by a surface alignment layer of FIG. 3C;

[0060] FIG. 4A, FIG. 4B, and FIG. 4C are schematic graphs illustrating the variation of transmission with polar direction for the polar control retarder of FIG. 2A using the twisted nematic polar control retarder of FIG. 3 for different points on the display device of FIG. 2A and operating at different voltages;

[0061] FIG. 5A is a perspective view of a polar control retarder that may be applied in FIG. 2A, comprising a twisted nematic switchable LC retarder and profile of voltage in the lateral direction across at least one region of the polar control retarder in a share mode of operation;

[0062] FIG. 5B is a side view of propagation of output light from a SLM through the optical stack of FIG. 5A;

[0063] FIG. 5C is a schematic graph illustrating the variation of transmission with polar direction for the polar control retarder of FIG. 16A with no voltage profile across the polar control retarder;

[0064] FIG. 5D is a schematic graph illustrating the variation of transmission with polar direction for a polar control retarder using the twisted nematic polar control retarder with additional polariser and display polariser that have electric vector transmission directions that are parallel operating in privacy mode;

[0065] FIG. 5E is a schematic graph illustrating the variation of transmission with polar direction for a polar control retarder using the twisted nematic polar control retarder with additional polariser and display polariser that have electric vector transmission directions that are parallel operating in share mode;

[0066] FIG. 6A is a schematic top view illustrating schematically the operation for a driver of a privacy display comprising the polar control retarder with a voltage profile across the polar control retarder and a head tracking system arranged to adjust the voltage across the polar control retarder in response to the measured location of the driver for the driver moving in the predetermined direction;

[0067] FIG. 6B is a schematic graph illustrating the variation in voltage profile applied to the polar control with respect to lateral position across the polar control retarder for two different driver lateral locations for the driver moving in the predetermined direction;

[0068] FIG. 6C is a schematic top view illustrating schematically the operation for a driver of a privacy display comprising the polar control retarder with a voltage profile across the polar control retarder and a head tracking system arranged to adjust the voltage across the polar control retarder in response to the measured location of the driver moving towards the display;

[0069] FIG. 6D is a schematic graph illustrating the variation in voltage profile applied to the polar control with respect to lateral position across the polar control retarder for two different driver lateral locations for the driver moving towards the display;

[0070] FIG. 6E is a schematic top view illustrating schematically the operation for a driver of a privacy display comprising the polar control retarder with a common voltage across the polar control retarder and a head tracking system arranged to adjust the common voltage across the polar control retarder in response to the measured location of the driver;

[0071] FIG. 6F is a schematic graph illustrating the variation in common voltage applied to the polar control for two different driver lateral locations;

[0072] FIG. 7A is a schematic top view of operation of a privacy display for a moving driver and a head tracking system arranged to minimise luminance to the driver;

[0073] FIG. 7B, FIG. 7C, and FIG. 7D are graphs illustrating the variation of transmission with polar direction for the polar control retarder of FIG. 2A using the twisted nematic polar control retarder of FIG. 3 operating with different voltage profiles across the polar control retarder;

[0074] FIG. 8A is a graph illustrating a simulated polar profile of the security level, S of the arrangement of FIG. 2A for an ambient illuminance measured in lux that is equal to the head-on display luminance measured in nits and for the polar profile of backlight luminance of FIG. 16A and polar profile of polar control retarder transmission of FIG. 4B;

[0075] FIG. 8B is a graph illustrating a simulated polar profile of the security level, S of the arrangement of FIG. 2A for an ambient illuminance measured in lux that is equal to the head-on display luminance measured in nits and for the polar profile of backlight luminance of FIG. 16A and polar profile of polar control retarder transmission of FIG. 4A;

[0076] FIG. 9A is a flow chart illustrating a method to control an observer-tracked polar control retarder;

[0077] FIG. 9B is a flow chart illustrating a method for manual control of an adjustable polar control retarder;

[0078] FIG. 10A is a schematic top view illustrating schematically the operation for a passenger of a privacy display comprising the polar control retarder with voltage profile of FIG. 2A;

[0079] FIG. 10B is a schematic top view illustrating schematically the operation for a driver of a privacy display comprising the polar control retarder with voltage profile of FIG. 2A;

[0080] FIG. 11A is a schematic top view illustrating schematically the operation for a passenger and a driver of a privacy display comprising the polar control retarder with a common voltage in the lateral direction across at least one region of the polar control retarder;

[0081] FIG. 11B is a schematic graph illustrating the variation of transmission across the display surface with polar direction for a polar control retarder with a common voltage across the polar control retarder;

[0082] FIG. 12A is a schematic diagram illustrating in front view a voltage profile across an area electrode comprising a view angle control arrangement for a privacy display;

[0083] FIG. 12B is a schematic diagram illustrating in front view a voltage profile across an area electrode comprising a view angle control arrangement using bus bars;

[0084] FIG. 12C is a schematic illustration of a circuit to produce a voltage profile in the x direction;

[0085] FIG. 12D is a schematic illustration of a circuit to produce a voltage profile in the x direction;

[0086] FIG. 13A is an illustration in perspective view of the opposing electrodes of a view angle control arrangement;

[0087] FIG. 13B is an illustration in perspective view of the opposing electrodes of a view angle control arrangement utilising bus bars for one electrode layer;

[0088] FIG. 13C is an illustration in perspective view of the opposing electrodes of a view angle control arrangement utilising bus bars for both electrodes;

[0089] FIG. 13D is a schematic graphical illustration of the different curved response functions;

[0090] FIG. 13E is an illustration in perspective view of the opposing electrodes of a view angle control arrangement a segmented electrode;

[0091] FIG. 13F is a schematic graphical illustration of a stepped response function with a series of voltage steps;

[0092] FIG. 13G is a schematic diagram illustrating in a perspective front view the opposing electrodes of a switchable liquid crystal retarder comprising corner bus bar contacts;

[0093] FIG. 14A is a schematic top view illustrating schematically the operation for a passenger of a privacy display comprising a backlight of FIG. 2A comprising a first waveguide comprising a first luminance profile;

[0094] FIG. 14B is a schematic top view illustrating the operation for a driver of a privacy display comprising a backlight of FIG. 2A comprising a second waveguide comprising a second luminance profile;

[0095] FIG. 15A is a schematic top view illustrating a curved privacy display comprising the polar control retarder with a voltage profile across the polar control retarder;

[0096] FIG. 15B is a schematic diagram illustrating a schematic top view of a pair of co-planar tiled display devices;

[0097] FIG. 15C is a schematic diagram illustrating a schematic top view of a pair of tilted planar tiled display devices;

[0098] FIG. 15D is a schematic diagram illustrating a schematic top view of a segmented curved display device comprising offset common points;

[0099] FIG. 15E is a schematic diagram illustrating a schematic top view of a segmented curved display device comprising different polar control retarder drive voltages;

[0100] FIG. 15F is a schematic diagram illustrating a schematic top view of a segmented curved display device comprising a planar part and a curve part;

[0101] FIG. 16A is a schematic graph illustrating the polar variation of luminance output for an alternative backlight of FIG. 2 primarily operating to direct light to the passenger;

[0102] FIG. 16B is a schematic graph illustrating the polar variation of luminance output for an

alternative backlight of FIG. 2 primarily operating to direct light to the driver;

[0103] FIG. 17A is a schematic graph illustrating the polar variation of luminance output for an alternative backlight of FIG. 2 operating to direct light to the passenger and to the driver;

[0104] FIG. 17B is a schematic graph illustrating the variation of relative luminance output at zero elevation for the alternative backlight profile of FIG. 5C operating to direct light to the passenger and driver;

[0105] FIG. 18 is a schematic graph illustrating a centre stack display for an automotive vehicle;

[0106] FIG. 19A is a schematic side perspective view of a display device providing uniformity for a display user wherein the polariser transmission directions of the display polarisers are different to those of FIG. 2A;

[0107] FIG. 19B is a front view of the stack of layers of the display device of FIG. 19A;

[0108] FIG. 20A, FIG. 20B, FIG. 20C, FIG. 20D, FIG. 20E, and FIG. 20F are graphs illustrating the variation of transmission with polar direction for various arrangements of liquid crystal polar control retarder, display polarisers and additional polarisers;

[0109] FIG. 21 is a schematic side perspective view of a display device comprising an emissive spatial light modulator and a polar control retarder and additional polariser arranged to receive light from the spatial light modulator;

[0110] FIG. 22 is a schematic side perspective view of an optical component comprising a polar control retarder comprising electrodes arranged to provide a variable profile of voltage in the lateral direction across at least one region of the polar control retarder;

[0111] FIG. 23A is a schematic side perspective view of a display device providing uniformity for a display user comprising first and second polar control retarders and a further additional polariser;

[0112] FIG. 23B is a schematic side perspective view of a display device comprising first and second polar control retarders with further additional polariser and second polar control retarder 300B arranged on the output side of the display output polariser;

[0113] FIG. 23C is schematic side perspective view of a display device comprising first and second polar control retarders with additional polariser and first polar control retarder arranged on the output side of the display output polariser and additional polariser and second polar control retarder arranged on the output side of the additional polariser;

[0114] FIG. 24A is a schematic side perspective view of a display device providing uniformity for a display user comprising a first polar control retarder with a profile of voltage and a second polar control retarder with a profile of alignment layer orientations;

[0115] FIG. 24B is schematic side perspective view of polar control retarders that may be applied in FIG. 24A, comprising a homeotropically and homogeneously aligned switchable LC retarder and a negative C-plate retarder wherein the alignment direction of the homeotropic alignment layer is common across the lateral direction and wherein the alignment direction of the homogeneous alignment layer varies across the lateral direction;

[0116] FIG. 24C is a schematic graph illustrating the variation of transmission with polar direction for the polar control retarder of FIG. 24B in a privacy mode of operation;

[0117] FIG. 24D is a schematic graph illustrating the variation of transmission with polar direction for the polar control retarder of FIG. 24B in a share mode of operation;

[0118] FIG. 25A is a schematic side perspective view of a display device providing uniformity for a display user comprising a first polar control retarder with a profile of voltage arranged between the backlight and transmissive spatial light modulator; and a reflective polariser, further polar control retarder and further additional polariser arranged to receive light from the spatial light modulator;

[0119] FIG. 25B is a schematic graph illustrating the variation of reflectivity with polar direction for the polar control retarder of FIG. 25A in a share mode of operation;

[0120] FIG. 25C is a graph illustrating a simulated polar profile of the security level, S of the arrangement of FIG. 25 for an ambient illuminance measured in lux that is the same as the head-on display luminance measured in nits and for the polar profile of backlight luminance of FIG. 16A

and polar profile of polar control retarder transmission of FIG. 24C and reflectivity of FIG. 25B;

[0121] FIG. 26A is a schematic diagram illustrating a side view of a switchable backlight comprising first and second waveguides, a rear reflector and an optical turning film and outputting light beams for passenger and driver nominal directions;

[0122] FIG. 26B is a schematic diagram illustrating a front perspective view of an optical turning film component for the backlight of FIG. 26A;

[0123] FIG. 26C is a schematic diagram illustrating a side view of an optical turning film component;

[0124] FIG. 27 is a schematic diagram illustrating a side view of a switchable backlight comprising a waveguide, first and second light sources at respective opposite input sides of the waveguide, a rear reflector and an optical turning film and outputting light beams for passenger and driver nominal directions;

[0125] FIG. 28A is a schematic diagram illustrating in side view operation of a backlight comprising a pupillating light turning film component;

[0126] FIG. 28B is a schematic diagram illustrating in front perspective view, operation of a pupillating light turning film component;

[0127] FIG. 29 is a schematic diagram illustrating a side view of a switchable backlight comprising a first waveguide, light turning film, a micro-louvre sheet and second waveguide;

[0128] FIG. 30 is a schematic diagram illustrating a side view of a switchable backlight comprising a first waveguide, prismatic recirculation films, a micro-louvre sheet and second waveguide;

[0129] FIG. 31 is a schematic diagram illustrating a side view of a switchable backlight comprising a mini-LED array, prismatic recirculation films, a micro-louvre sheet and second waveguide;

[0130] FIG. 32A is a schematic diagram illustrating a perspective side view of a steerable backlight comprising a stepped waveguide, and addressable light source array;

[0131] FIG. 32B is a schematic diagram illustrating a side view of a steerable backlight comprising a stepped waveguide, and addressable light source array and rear reflector; and

[0132] FIG. 32C is a schematic diagram illustrating a perspective rear view of a steerable backlight comprising a stepped waveguide, addressable light source array to illuminate a driver and passenger.

DETAILED DESCRIPTION

[0133] Terms related to optical retarders for the purposes of the present disclosure will now be described.

[0134] In a layer comprising a uniaxial birefringent material there is a direction governing the optical anisotropy whereas all directions perpendicular to it (or at a given angle to it) have equivalent birefringence.

[0135] The optical axis of an optical retarder refers to the direction of propagation of a light ray in the uniaxial birefringent material in which no birefringence is experienced. This is different from the optical axis of an optical system which may for example be parallel to a line of symmetry or normal to a display surface along which a principal ray propagates.

[0136] For light propagating in a direction orthogonal to the optical axis, the optical axis is the slow axis when linearly polarized light with an electric vector direction parallel to the slow axis travels at the slowest speed. The slow axis direction is the direction with the highest refractive index at the design wavelength. Similarly the fast axis direction is the direction with the lowest refractive index at the design wavelength.

[0137] For positive dielectric anisotropy uniaxial birefringent materials the slow axis direction is the extraordinary axis of the birefringent material. For negative dielectric anisotropy uniaxial birefringent materials the fast axis direction is the extraordinary axis of the birefringent material.

[0138] The terms half a wavelength and quarter a wavelength refer to the operation of a retarder for a design wavelength $\lambda_{sub.0}$ that may typically be between 500 nm and 570 nm. In the present illustrative embodiments exemplary retardance values are provided for a wavelength of 550 nm

unless otherwise specified.

[0139] The retarder provides a phase shift between two perpendicular polarization components of the light wave incident thereon and is characterized by the amount of relative phase, Γ , that it imparts on the two polarization components; which is related to the birefringence Δn and the thickness d of the retarder by

$$\Gamma = 2\pi \Delta n d / \lambda_0 \quad \text{eqn. 1}$$

[0140] In eqn. 1, Δn is defined as the difference between the extraordinary and the ordinary index of refraction, i.e.

$$\Delta n = n_e - n_o \quad \text{eqn. 2}$$

[0141] For a half-wave retarder, the relationship between d , Δn , and λ_0 is chosen so that the phase shift between polarization components is $\Gamma = \pi$. For a quarter-wave retarder, the relationship between d , Δn , and λ_0 is chosen so that the phase shift between polarization components is $\Gamma = \pi/2$.

[0142] The term half-wave retarder herein typically refers to light propagating normal to the retarder and normal to the spatial light modulator.

[0143] Some aspects of the propagation of light rays through a transparent retarder between a pair of polarisers will now be described.

[0144] The state of polarisation (SOP) of a light ray is described by the relative amplitude and phase shift between any two orthogonal polarization components. Transparent retarders do not alter the relative amplitudes of these orthogonal polarisation components but act only on their relative phase. Providing a net phase shift between the orthogonal polarisation components alters the SOP whereas maintaining net relative phase preserves the SOP. In the current description, the SOP may be termed the polarisation state.

[0145] A linear SOP has a polarisation component with a non-zero amplitude and an orthogonal polarisation component which has zero amplitude.

[0146] A linear polariser transmits a unique linear SOP that has a linear polarisation component parallel to the electric vector transmission direction of the linear polariser and attenuates light with a different SOP.

[0147] Absorbing polarisers are polarisers that absorb one polarisation component of incident light and transmit a second orthogonal polarisation component. Examples of absorbing linear polarisers are dichroic polarisers.

[0148] Reflective polarisers are polarisers that reflect one polarisation component of incident light and transmit a second orthogonal polarisation component. Examples of reflective polarisers that are linear polarisers are multilayer polymeric film stacks such as DBEF™ or APF™ from 3M Corporation, or wire grid polarisers such as ProFlux™ from Moxtek. Reflective linear polarisers may further comprise cholesteric reflective materials and a quarter waveplate arranged in series.

[0149] A retarder arranged between a linear polariser and a parallel linear analysing polariser that introduces no relative net phase shift provides full transmission of the light other than residual absorption within the linear polariser.

[0150] A retarder that provides a relative net phase shift between orthogonal polarisation components changes the SOP and provides attenuation at the analysing polariser.

[0151] In the present disclosure an 'A-plate' refers to an optical retarder utilizing a layer of birefringent material with its optical axis parallel to the plane of the layer.

[0152] A 'positive A-plate' refers to positively birefringent A-plates, i.e. A-plates with a positive Δn .

[0153] In the present disclosure a 'C-plate' refers to an optical retarder utilizing a layer of birefringent material with its optical axis perpendicular to the plane of the layer. A 'positive C-plate' refers to positively birefringent C-plates, i.e. C-plates with a positive Δn . A 'negative C-plate' refers to negatively birefringent C-plates, i.e. C-plates with a negative Δn .

[0154] 'O-plate' refers to an optical retarder utilizing a layer of birefringent material with its optical axis having a component parallel to the plane of the layer and a component perpendicular to the plane of the layer. A 'positive O-plate' refers to positively birefringent O-plates, i.e. O-plates with a positive Δn .

[0155] Achromatic retarders may be provided wherein the material of the retarder is provided with a retardance $\Delta n \cdot d$ that varies with wavelength λ as

[00003] $n \cdot d / \lambda^3 = \text{eqn. 3}$

[0156] where κ is substantially a constant.

[0157] Examples of suitable materials include modified polycarbonates from Teijin Films.

Achromatic retarders may be provided in the present embodiments to advantageously minimise colour changes between polar angular viewing directions which have low luminance reduction and polar angular viewing directions which have increased luminance reductions as will be described below.

[0158] Various other terms used in the present disclosure related to retarders and to liquid crystals will now be described.

[0159] A liquid crystal cell has a retardance given by $\Delta n \cdot d$ where Δn is the birefringence of the liquid crystal material in the liquid crystal cell and d is the thickness of the liquid crystal cell, independent of the alignment of the liquid crystal material in the liquid crystal cell.

[0160] Homogeneous alignment refers to the alignment of liquid crystals in switchable liquid crystal displays where molecules align substantially parallel to a substrate. Homogeneous alignment is sometimes referred to as planar alignment. Homogeneous alignment may typically be provided with a small pre-tilt such as 2 degrees, so that the molecules at the surfaces of the alignment layers of the liquid crystal cell are slightly inclined as will be described below. Pretilt is arranged to minimise degeneracies in switching of cells.

[0161] In the present disclosure, homeotropic alignment is the state in which rod-like liquid crystalline molecules align substantially perpendicularly to the substrate. In discotic liquid crystals homeotropic alignment is defined as the state in which an axis of the column structure, which is formed by disc-like liquid crystalline molecules, aligns perpendicularly to a surface. In homeotropic alignment, pretilt is the tilt angle of the molecules that are close to the alignment layer and is typically close to 90 degrees and for example may be 88 degrees.

[0162] In a twisted liquid crystal layer, a twisted configuration (also known as a helical structure or helix) of nematic liquid crystal molecules is provided. The twist may be achieved by means of a non-parallel alignment of alignment layers. Further, cholesteric dopants may be added to the liquid crystal material to break degeneracy of the twist direction (clockwise or anti-clockwise) and to further control the pitch of the twist in the relaxed (typically undriven) state. A super-twisted liquid crystal layer has a twist of greater than 180 degrees. A twisted nematic layer used in spatial light modulators typically has a twist of 90 degrees.

[0163] Liquid crystal molecules with positive dielectric anisotropy are switched from a homogeneous alignment (such as an A-plate retarder orientation) to a homeotropic alignment (such as a C-plate or O-plate retarder orientation) by means of an applied electric field.

[0164] Liquid crystal molecules with negative dielectric anisotropy are switched from a homeotropic alignment (such as a C-plate or O-plate retarder orientation) to a homogeneous alignment (such as an A-plate retarder orientation) by means of an applied electric field.

[0165] Rod-like molecules have a positive birefringence so that $n_{\text{sub.e}} > n_{\text{sub.o}}$ as described in eqn. 2. Discotic molecules have negative birefringence so that $n_{\text{sub.e}} < n_{\text{sub.o}}$.

[0166] Positive retarders such as A-plates, positive O-plates and positive C-plates may typically be provided by stretched films or rod-like liquid crystal molecules. Negative retarders such as negative C-plates may be provided by stretched films or discotic-like liquid crystal molecules.

[0167] Parallel liquid crystal cell alignment refers to the alignment direction of homogeneous alignment layers being parallel or more typically antiparallel. In the case of pre-tilted homeotropic

alignment, the alignment layers may have components that are substantially parallel or antiparallel. Hybrid aligned liquid crystal cells may have one homogeneous alignment layer and one homeotropic alignment layer. Twisted liquid crystal cells may be provided by alignment layers that do not have parallel alignment, for example oriented at 90 degrees to each other.

[0168] Transmissive spatial light modulators may further comprise retarders between the input display polariser and the output display polariser for example as disclosed in U.S. Pat. No. 8,237,876, which is herein incorporated by reference in its entirety. Such retarders (not shown) are in a different place to the passive retarders of the present embodiments. Such retarders compensate for contrast degradations for off-axis viewing locations, which is a different effect to the luminance reduction for off-axis viewing positions of the present embodiments.

[0169] A private mode of operation of a display is one in which an observer sees a low contrast sensitivity such that an image is not clearly visible. Contrast sensitivity is a measure of the ability to discern between luminances of different levels in a static image. Inverse contrast sensitivity may be used as a measure of visual security, in that a high visual security level (VSL) corresponds to low image visibility.

[0170] For a privacy display providing an image to an observer, visual security may be given as:

[00004] $V = (Y + R) / (Y - K)$ eqn. 4

[0171] where V is the visual security level (VSL), Y is the luminance of the white state of the display at a snoop viewing angle, K is the luminance of the black state of the display at the snoop viewing angle and R is the luminance of reflected light from the display.

[0172] Panel contrast ratio is given as:

[00005] $C = Y / K$ eqn. 5

[0173] so the visual security level may be further given as:

[00006] $V = (P \cdot Y_{\max} + I \cdot \rho) / (P \cdot (Y_{\max} - Y_{\min}) / C)$ eqn. 6

[0174] where: $Y_{\max}$ is the maximum luminance of the display; P is the off-axis relative luminance typically defined as the ratio of luminance at the snoop angle to the maximum luminance $Y_{\max}$; C is the image contrast ratio; ρ is the surface reflectivity; and I is the illuminance. The units of $Y_{\max}$ are the units of I divided by solid angle in units of steradian.

[0175] The luminance of a display varies with angle and so the maximum luminance of the display $Y_{\max}$ occurs at a particular angle that depends on the configuration of the display.

[0176] In many displays, the maximum luminance $Y_{\max}$ occurs head-on, i.e. normal to the display. Any display device disclosed herein may be arranged to have a maximum luminance $Y_{\max}$ that occurs head-on, in which case references to the maximum luminance of the display device $Y_{\max}$ may be replaced by references to the luminance normal to the display device.

[0177] Alternatively, any display described herein may be arranged to have a maximum luminance $Y_{\max}$ that occurs at a polar angle to the normal to the display device that is greater than 0°. By way of example, the maximum luminance $Y_{\max}$ may occur at a non-zero polar angle and at an azimuth angle that has for example zero lateral angle so that the maximum luminance is for an on-axis user that is looking down on to the display device. The polar angle may for example be 10 degrees and the azimuthal angle may be the northerly direction (90 degrees anti-clockwise from easterly direction). The viewer may therefore desirably see a high luminance at typical non-normal viewing angles.

[0178] The off-axis relative luminance, P is sometimes referred to as the privacy level. However, such privacy level P describes relative luminance of a display at a given polar angle compared to head-on luminance, and in fact is not a measure of privacy appearance.

[0179] The illuminance, I is the luminous flux per unit area that is incident on the display and reflected from the display towards the observer location. For Lambertian illuminance, and for displays with a Lambertian front diffuser, illuminance I is invariant with polar and azimuthal angles. For arrangements with a display with non-Lambertian front diffusion arranged in an

environment with directional (non-Lambertian) ambient light, illuminance I varies with polar and azimuthal angle of observation.

[0180] Thus in a perfectly dark environment, a high contrast display has VSL of approximately 1.0. As ambient illuminance increases, the perceived image contrast degrades, VSL increases and a private image is perceived.

[0181] For typical liquid crystal displays the panel contrast C is above 100:1 for almost all viewing angles, allowing the visual security level to be approximated to:

[00007] $V = 1 + I / (P \cdot Y_{\max})$ eqn. 7

[0182] In the present embodiments, in addition to the exemplary definition of eqn. 4, other measurements of visual security level, V may be provided, for example to include the effect on image visibility to a snooper of snooper location, image contrast, image colour and white point and subtended image feature size. Thus the visual security level may be a measure of the degree of privacy of the display but may not be restricted to the parameter V .

[0183] The perceptual image security may be determined from the logarithmic response of the eye, such that

[00008] $S = \log_{10}(V)$ eqn. 8

[0184] Desirable limits for S were determined in the following manner. In a first step a privacy display device was provided. Measurements of the variation of privacy level, $P(\theta)$ of the display device with polar viewing angle and variation of reflectivity $\rho(\theta)$ of the display device with polar viewing angle were made using photopic measurement equipment. A light source such as a substantially uniform luminance light box was arranged to provide illumination from an illuminated region that was arranged to illuminate the privacy display device along an incident direction for reflection to a viewer position at a polar angle of greater than 0° to the normal to the display device. The variation $I(\theta)$ of illuminance of a substantially Lambertian emitting lightbox with polar viewing angle was determined by and measuring the variation of recorded reflective luminance with polar viewing angle taking into account the variation of reflectivity $\rho(\theta)$. The measurements of $P(\theta)$, $\rho(\theta)$ and $I(\theta)$ were used to determine the variation of Security Factor $S(\theta)$ with polar viewing angle along the zero elevation axis.

[0185] In a second step a series of high contrast images were provided on the privacy display including (i) small text images with maximum font height 3 mm, (ii) large text images with maximum font height 30 mm and (iii) moving images.

[0186] In a third step each observer (with eyesight correction for viewing at 1000 mm where appropriate) viewed each of the images from a distance of 1000 mm, and adjusted their polar angle of viewing at zero elevation until image invisibility was achieved for one eye from a position near on the display at or close to the centre-line of the display. The polar location of the observer's eye was recorded. From the relationship $S(\theta)$, the security factor at said polar location was determined. The measurement was repeated for the different images, for various display luminance $Y_{\text{sub.max}}$, different lightbox illuminance $I(\theta=0)$, for different background lighting conditions and for different observers.

[0187] From the above measurements $S < 1.0$ provides low or no visual security, $1.0 \leq S < 1.5$ provides visual security that is dependent on the contrast, spatial frequency and temporal frequency of image content, $1.5 \leq S < 1.8$ provides acceptable image invisibility (that is no image contrast is observable) for most images and most observers and $S \geq 1.8$ provides full image invisibility, independent of image content for all observers.

[0188] In practical display devices, this means that it is desirable to provide a value of S for an off-axis viewer who is a snooper that meets the relationship $S \geq S_{\text{sub.min}}$, where: $S_{\text{sub.min}}$ has a value of 1.0 or more to achieve the effect that the off-axis viewer cannot perceive the displayed image; $S_{\text{sub.min}}$ has a value of 1.5 or more to achieve the effect that the displayed image is invisible, i.e. the viewer cannot perceive even that an image is being displayed, for most images and most

observers; or $S_{\text{sub.min}}$ has a value of 1.8 or more to achieve the effect that the displayed image is invisible independent of image content for all observers.

[0189] In comparison to privacy displays, desirably wide angle displays are easily observed in standard ambient illuminance conditions. One measure of image visibility is given by the contrast sensitivity such as the Michelson contrast which is given by:

[00009] $M = (I_{\text{max}} - I_{\text{min}}) / (I_{\text{max}} + I_{\text{min}})$ eqn. 9 andso: eqn. 10

$$M = ((Y + R) - (K + R)) / ((Y + R) + (K + R)) = (Y - K) / (Y + K + 2 \cdot \text{Math. } R)$$

[0190] Thus the visual security level (VSL), V is equivalent (but not identical to) $1/M$. In the present discussion, for a given off-axis relative luminance, P the wide angle image visibility, W is approximated as

[00010] $W = 1 / V = 1 / (1 + I \cdot \text{Math. } / (\cdot \text{Math. } P \cdot \text{Math. } Y_{\text{max}}))$ eqn. 11

[0191] The above discussion focusses on reducing visibility of the displayed image to an off-axis viewer who is a snooper, but similar considerations apply to visibility of the displayed image to the intended user of the display device who is typically on-axis. In this case, decrease of the level of the visual security level (VSL) V corresponds to an increase in the visibility of the image to the viewer. During observation $S < 0.1$ may provide acceptable visibility of the displayed image. In practical display devices, this means that it is desirable to provide a value of S for an on-axis viewer who is the intended user of the display device that meets the relationship $S \leq S_{\text{sub.max}}$, where $S_{\text{sub.max}}$ has a value of 0.1.

[0192] In the present discussion the colour variation $\Delta\epsilon$ of an output colour ($u_{\text{sub.w'}} + \Delta u'$, $v_{\text{sub.w'}} + \Delta v'$) from a desirable white point ($u_{\text{sub.w'}}$, $v_{\text{sub.w'}}$) may be determined by the CIELUV colour difference metric, assuming a typical display spectral illuminant and is given by:

[00011] $= (u'^2 + v'^2)^{1/2}$ eqn. 12

[0193] Switchable directional display apparatuses for use in privacy display for example and comprising plural retarders arranged between a display polariser and an additional polariser are described in U.S. Pat. Nos. 11,092,851, 10,948,648, and WIPO Publ. No. 2019-055755 (U.S. Pat. No. 11,099,433), all of which are herein incorporated by reference in their entireties. Directional display apparatuses further comprising reflective polarisers arranged between the display polariser and retarders are described in U.S. Pat. No. 10,976,578 and in U.S. Pat. No. 10,802,356, both of which are herein incorporated by reference in their entireties. Directional display polarisers comprising passive retarders arranged between a display polariser and an additional polariser are described U.S. Pat. No. 11,327,358, which is herein incorporated by reference in its entirety.

[0194] Curvature is a property of a line that is curved and for the present disclosure is the inverse radius of curvature. A planar surface has a curvature of zero.

[0195] The structure and operation of various directional display devices will now be described. In this description, common elements have common reference numerals. It is noted that the disclosure relating to any element applies to each device in which the same or corresponding element is provided. Accordingly, for brevity such disclosure is not repeated.

[0196] A switchable privacy display such as for a vehicle will now be described.

[0197] FIG. 1A is a schematic diagram illustrating a schematic top view of a privacy display device **100** for use by a passenger in a vehicle. Vehicles may include the automotive vehicle **650** of FIG. 1A or trains, boats, and airplanes for example.

[0198] Display device **100** is arranged in a passenger information display (PID) location, (on the right-hand side of the vehicle for Left-Hand Drive), with light rays **445**, **447** output to the user **45** and user **47** respectively. In right-hand drive vehicles, the directions of light deflection referred to hereinbelow are typically reflected about a vertical axis, that is the lateral direction is reversed.

[0199] In a first mode of operation that is the privacy mode the display device **100** is arranged for viewing by the front passenger **45** near to an on-axis **199** location, and to inhibit viewing by the driver **47**. In operation in privacy mode, the light rays **447** may be directed towards a common off-

axis point **427** in front of the display device **100**. In privacy mode, the light rays **447** may represent the direction for minimum luminance from each point on the display device **100**.

[0200] In the present disclosure, pupillation refers to the optical output of the display providing at least one common point such as off-axis point **427** from which rays from each at least part of the display device **100** are directed with substantially similar transmission, or luminance. An observer at a pupil such as point **427** may see a substantially uniform luminance or transmission from across the at least part of the display device **100**. As will be described hereinbelow, pupillation of various components in the display **100** may advantageously achieve increased luminance uniformity and increased uniformity of security factor.

[0201] It is desirable that the passenger **45** may view information such as entertainment without the image causing distraction to the driver **47**, that is the privacy mode refers to a low driver distraction mode. This mode is in comparison with a mode in which the passenger display turns off when the vehicle is in motion to prevent driver distraction. More specifically to minimise the visibility to the driver **47** of distracting images at both the nominal driver position and when the driver leans across towards the display while driving, it is desirable to maximise the security factor S at angles α from the optical axis **199** of greater than 30° and preferably greater than 25° in the direction from the optical axis **199** towards the driver **47**. Further it is desirable to achieve a high security factor, S for polar angles at least at angles β from the optical axis **199**.

[0202] Optional observer position location viewer tracking system **200** may comprise a camera with collection cone angle **201** and is provided to detect the location of at least the driver **47** during driver **47** movement **540**. The observer tracking system may optionally comprise more than one camera, for example one to detect the driver **47** and one to detect the passenger **45**. Viewer tracking system **200** may further comprise for example an image processing means to determine the location of an observer in a collected image. The image processing means may optionally detect the gaze direction of the driver **47** or passenger **45** as well as the head location. An additional function where the display brightness is reduced or turned off when the gaze is directed towards the display may be provided.

[0203] Further in a low stray light function of the privacy mode, it may be desirable to provide an image to the passenger **45** with desirable luminance while reducing the luminance to reflecting and scattering surfaces within the vehicle. Advantageously the brightness of internal surfaces of the vehicle **650** may be reduced during night-time operation, reducing driver distraction. Further, increased area displays may be provided while maintaining desirably low levels of stray illumination within the vehicle **650** cabin.

[0204] In a second mode that is the share mode, the display device **100** is arranged for viewing by driver **47** in an off-axis location. Such use may be for occasions when viewing the display content is safe such as when the vehicle is stationary, or the content is appropriate such as map or instrument data.

[0205] Further it would be desirable to achieve high image uniformity to the passenger **45** and high uniformity of security factor, S to the driver **47**.

[0206] FIG. **1B** is a schematic diagram illustrating a schematic top view of a privacy display device **100** for use in off-axis use applications. Features of the embodiment of FIG. **1B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0207] The present disclosure refers commonly to passenger **45** or driver **47** for an automotive vehicle **650** of FIG. **1A**. The present embodiments hereinbelow are suited to applications with a preferred side on which privacy may be provided. The display devices **100** herein may be used in other applications, in which case user **47** or snooper **47** and user **45** may be equally substituted.

[0208] The display device **100** may be provided for applications including but not limited to point-of-sale, or medical monitors. Considering the alternative embodiment of FIG. **1B**, such a display device **100** may be viewed on the right side by a user **49** who is a customer or medical practitioner,

centrally by a user **45** who is a customer or patient and shielded from potential viewers **47** who may be other members of the public on the opposite side of the optical axis **199** to the user **49**.

[0209] In a share mode of operation the display **100** may be used to provide advertising information over a wide polar region, or to provide medical information to multiple display users for example. As described hereinbelow, the embodiments of FIGS. **23A-C** for example may be used to provide switching of the side of the display on which the user **47** is located.

[0210] The embodiment of FIG. **1B** may be provided with location viewer tracking system **200** and optionally or alternatively may comprise manual actuator **202** to provide user input indicating a direction of minimum light transmission of a view angle control arrangement **310** as will be described further hereinbelow.

[0211] An illustrative structure that can achieve the desirable characteristics of the display device **100** of FIG. **1A** will now be described.

[0212] FIG. **2A** is a schematic side perspective view of a display device **100** providing uniformity in reduction of luminance in directions; and FIG. **2B** is a schematic side view of an optical stacking for the display device of FIG. **2A**. Features of the arrangement of FIG. **2A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features

[0213] FIG. **2A** illustrates the display device **100**, for use in ambient illumination **604**, which comprises a spatial light modulator (SLM) **48** arranged to output light **400**. The SLM **48** comprises an input polariser **210** arranged on the input side of the SLM **48** and an output polariser **218**, arranged on the output side of the SLM **48**, the input polariser **210** and the output polariser **218** being the two display polarisers of the SLM **48**. The input polariser **210** and the output polariser **218** are each linear polarisers.

[0214] The display device **100** also comprises an additional polariser **318** arranged on the input side of the input polariser **210**, that is the same side as the input polariser **210**. The additional polariser **318** is a linear polariser. Typical polarisers **210**, **218**, **318** may be polarisers such as dichroic polarisers.

[0215] In the present disclosure, the SLM **48** may comprise a liquid crystal display comprising substrates **212**, **216**, liquid crystal layer **214** and red, green and blue pixels **220**, **222**, **224**. The output polariser **218** may be arranged to provide high extinction ratio for light from the pixels **220**, **222**, **224** of the SLM **48**.

[0216] The display device **100** also comprises at least one polar control retarder **300** which comprises a switchable liquid crystal retarder **301** comprising a layer **314** of liquid crystal material, and substrates **312**, **316** arranged between the input polariser **210** and the additional polariser **318**. The substrates **312**, **316** of the switchable liquid crystal retarder **301** respectively comprise electrodes **320A**, **320B** arranged to provide a voltage across the layer **314** of liquid crystal material **414** for controlling the layer **314**. A control system **352** is arranged to control the voltage applied, by a voltage driver **350**, across the electrodes **320A**, **320B** of the switchable liquid crystal retarder **301**.

[0217] Electrodes **320A**, **320B** are connected to control system **500** by means of at least one electrical bus bar contact **321** and voltage driver system **352**. Electrodes **320A**, **320B** are arranged to provide switching voltage across the layer **314** of liquid crystal material **414**. Voltage is applied with voltage profile **322** that varies across the lateral direction of the liquid crystal layer **314** as will be described further hereinbelow.

[0218] In typical use, for switching between a share mode and a privacy mode, the layer **314** of liquid crystal material **414** is switchable between two states. The first state being a share mode so that the display device **100** may be used by multiple users, the second state being a privacy mode for use by a primary user with minimal visibility by snoopers. The switching may be by means of a voltage being applied across the electrodes **320A**, **320B**.

[0219] In the embodiment of FIG. **2A**, the display device **100** further comprises a backlight **20**

arranged to illuminate the SLM **48**. The SLM **48** comprises a transmissive SLM **48** arranged to receive and spatially modulate the output light from the backlight **20**, the display device **100** further comprises a backlight **20** arranged to illuminate the SLM **48**, and said display polariser **210** is an input display polariser **210** arranged on the input side of the SLM **48**.

[0220] Backlight **20** may be arranged to illuminate the SLM **48**, thereby providing a transmissive SLM **48** and may comprise input light sources **15A**, **15B**, waveguides **1A**, **1B**, rear reflector **3** and optical stack **5** comprising diffusers, light turning films and other known optical backlight structures. Plural light sources **15A**, **15B** are shown by way of non-limitative example, but in general there may any number of one or more light sources **15A**, **15B**.

[0221] In operation for illumination of passenger **45**, most generally light is directed along ray bundle **445** output from the display device **100**. Light rays **444A** from the first light sources **15A** are directed into light cones **460L**, **460C**, **460R** after transmission at facets **53A** and reflection at facets **53B** of light turning film component **50** and after transmission through the polar control retarder **300** that may be arranged in a privacy or share mode of operation.

[0222] In operation for illumination of driver **47**, most generally light is directed along ray bundle **447** output from the display device **100**. Light rays **444B** from the second light sources **15B** are directed into light cones **462L**, **462C**, **462R** after transmission at facets **53B** and reflection at facets **53A** of light turning film component **50** and after transmission through the polar control retarder **300** that is typically arranged in a share mode of operation.

[0223] Asymmetric diffusers, that may comprise asymmetric surface relief features for example, may be provided in the optical stack **5** with increased diffusion in the elevation direction in comparison to the lateral direction. Advantageously image uniformity may be increased.

[0224] In the alternative embodiment of FIG. 2B, an optical stacking for the arrangement of FIG. 2A is illustrated. Air gap **57** is provided between spatial light modulator **48** and substrate **316** of the polar control retarder **300**. Advantageously yield may be improved in manufacture. In an alternative embodiment the substrate **316** may be bonded to the polariser **210** so that reflection losses are advantageously reduced.

[0225] Light turning film **50** may be bonded to the additional polariser **318** by means of adhesive layer **51**. In operation, reflections from the top surface of the light turning film **50** are reduced in comparison to arrangements where the adhesive layer **51** is replaced by an air gap. Such an arrangement achieves reduced stray light directed into regions between the driver **47** and passenger **45** in the illustrative arrangement of FIG. 1A for example. Advantageously distraction to driver **47** may be reduced.

[0226] Polar control retarder **300** may comprise liquid crystal material **414** arranged between alignment layer **317A**, **317B** arranged on transparent electrodes **320A**, **320B** and arranged to provide alignment of liquid crystal material **414**. Bus bar **323** may be provided on at least one edge of the electrode **320B** to achieve substantially uniform electrical connectivity to the transparent electrode **320B**. The electrodes **320A** and **320B** may have different sheet resistance (ohms per square) conductors (such as ITO or silver nanowire). The higher sheet resistance and edge bus bars **321L**, **321R** may be used to provide a larger range profile **322**. The opposing electrode **320B** for example may have a lower sheet resistance in order to provide a sufficiently uniform voltage across it without using full bus bars **323**. In the present embodiments, the electrodes **321**, **323** may be provided on different ones of either of the rearmost and forwardmost electrode **320A**, **320B**.

[0227] FIG. 2C is a front view of the stack of some of the layers of the display device **100** of FIG. 2A. Features of the arrangement of FIG. 2C not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features. In FIG. 2C and other schematic diagrams below, some layers of the optical stack are omitted for clarity. For example, the switchable liquid crystal retarder **301** is shown omitting the substrates **312**, **316**.

[0228] The surface alignment layer **419B** closest to the display polariser **210** has a pretilt having a

pretilt direction **417Bp** with a component in the plane of the layer **314** of liquid crystal material **414** that is parallel or perpendicular to an electric vector transmission direction **211** of the display polarisers **210**, **218**, and the surface alignment layer **419A** closest to the additional polariser **318** has a pretilt having a pretilt direction **417Ap** with a component in the plane of the layer **314** of liquid crystal material **414** that is parallel or perpendicular to an electric vector transmission direction **319** of the additional polariser **318**. In the illustrative embodiment of FIG. 2C, the pretilt direction **417Ap** is parallel to the electric vector transmission direction **319**, and the pretilt direction **417Bp** is parallel to the electric vector transmission direction **211**.

[0229] The electrodes **320A**, **320B**, which are optically transmissive comprising a material such as ITO, are arranged across the layer **314** to control the liquid crystal material **414** and thereby control the liquid crystal retarder **301**. The layer **314** of liquid crystal material **414** is switchable by means of adjusting the voltage being applied to the electrodes **320A**, **320B**. The electrodes **320A**, **320B** are on opposite sides of the layer **314** of liquid crystal material **414** and may for example be indium-tin-oxide (ITO) electrodes.

[0230] The alignment layers **419A**, **419B** may be formed between electrodes **320A**, **320B** and the liquid crystal material **414** of the layer **314**.

[0231] The polar control retarder **301** further comprises two surface alignment layers **419A**, **419B** disposed adjacent to the layer **314** of liquid crystal material **414** and on opposite sides thereof. Each of the surface alignment layers **419A**, **419B** is arranged to provide alignment in the adjacent liquid crystal material **414** with an in-plane component **417Ap**, **417Bp** respectively that is in the plane of the layer **314** of liquid crystal material **414**.

[0232] A polar control retarder **300** wherein the SLM **48** is a twisted nematic (TN) SLM **48** may be arranged with the properties of TABLE 1 for example. When viewed from the front of the display device **100**, the optical axis direction **412** of the liquid crystal material **414** at the first alignment layer **419A** of the polar control retarder **300** has an in-plane component **417Ap** with an anti-clockwise rotation angle $\theta_{\text{sub.A}}$ from easterly direction of 135° . Similarly, the optical axis direction **417** of the liquid crystal material **414** at the second alignment layer **419B** has an in-plane component **417Bp** with an anti-clockwise rotation angle $\theta_{\text{sub.B}}$ from easterly direction of 45° .

TABLE-US-00001 TABLE 1 In-plane In-plane Active LC retarder 301 rotation rotation Alignment Item angle angle Twist layers Pretilt $\Delta n \cdot d$ V_{sub.C} 318 319, $\phi_{\text{sub.A}}$ 135° 314 417Ap, $\theta_{\text{sub.A}}$ 135° 90° Homogeneous 2° 500 nm 1.45 V 417Bp, $\theta_{\text{sub.B}}$ 45° Homogeneous 2° 210 211, $\phi_{\text{sub.B}}$ 45° 218 219, $\phi_{\text{sub.B'}}$ 135°

[0233] The liquid crystal material **414** has a positive dielectric anisotropy such that the optical axis of molecules of the liquid crystal material **414** lie at the pretilt angle to the plane of the layer **314** across the layer when no voltage is applied; and the molecules that are not near the alignment layers **419A**, **419B** are tilted towards the normal of the layer **314** when a voltage is applied. A wide polar angle of high transmission may be achieved advantageously with low power consumption for share mode operation.

[0234] The additional polariser **318** and display polariser **210** are illustrated as having respective electric vector transmission directions **319**, **211** that are crossed, that is they are at 90° to each other. In other embodiments including those described hereinbelow, the polarisers may be parallel or may be inclined to angles that are not parallel or crossed. Polar profiles of transmission may advantageously be modified to achieve desirable viewing characteristics for driver **47** or passenger **45**.

[0235] It would be desirable to provide high luminance uniformity to passenger **45** and high uniformity of security factor to driver **47** across at least the lateral direction of the display device **100**. At locations that are between the alignment layers **419A**, **419B**, and may vary across the lateral direction (x-axis) of the display, the optical axis direction **412** of the liquid crystal material **414** varies as will now be described.

[0236] FIG. 3A is a perspective view of the polar control retarder **300** of FIG. 2C, comprising a

twisted nematic switchable LC retarder **301** and voltage profile **322** in the lateral direction (x-axis) across at least one region of the polar control retarder **300** in a privacy mode of operation. Features of the arrangement of FIG. **3A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0237] View angle control arrangement **310** comprises: additional polariser **318** arranged on the same side of the SLM **48** as the display polariser **210**, the additional polariser **318** being a linear polariser; and at least one polar control retarder **300** arranged between the display polariser **210** and the additional polariser **318**.

[0238] The at least one polar control retarder **300** comprises a switchable liquid crystal retarder **301** comprising a layer **314** of liquid crystal material **414**; two surface alignment layers **419A**, **419B** disposed adjacent to the layer **314** of liquid crystal material **414** and on opposite sides thereof. The surface alignment layers **419A**, **419B** are each arranged to provide homogenous alignment in the adjacent liquid crystal material **414**, the liquid crystal material **414** being provided with a twist.

[0239] In other words, the switchable liquid crystal retarder **301** comprises two surface alignment layers **419A**, **419B** disposed adjacent to the layer **314** of liquid crystal material **414** and on opposite sides thereof and each arranged to provide homogeneous alignment in the adjacent liquid crystal material **414**.

[0240] FIG. **3A** illustrates that a cholesteric dopant **411** is provided in the layer of liquid crystal material **414** to minimise degeneracy in twist direction of the liquid crystal material **414** across the liquid crystal layer **314**. Twist may further be achieved by the different alignment directions **417Ap**, **417Bp**. Rotation angles different to 90 degrees, for example 80 degrees may achieve non-degenerate modes without the use of dopant **411**. Advantageously complexity and cost is reduced.

[0241] In the embodiment of FIG. **3A**, the electric vector transmission direction **319** of the additional polariser **318** is oriented at a non-zero angle with respect to the electric vector transmission direction **211** of the display polariser **210** and the non-zero angle is the same as the twist. Further arrangements of polarisers, electric vector transmission directions **319**, **211** and alignment layer orientations **417Ap**, **417Bp** will be described hereinbelow with respect to for example TABLE 5 wherein the non-zero angle may be different to the twist.

[0242] First and second electrode arrangements **320A**, **320B** are disposed on opposite sides of the layer **314** of liquid crystal material **414**, wherein first and second electrode arrangements **320A**, **320B** are arranged to provide an electric field **370** perpendicular to the layer **314** of liquid crystal material **414**. For explanatory purposes, the magnitude of the electric field **370** is illustrated with field strength indicated by field lines with differing densities across a predetermined axis.

[0243] In the embodiment of FIG. **2A** and FIG. **3A**, the predetermined axis is illustrated as the x-axis, that is the axis that runs from left to right across the display. The predetermined axis may be the nominal axis along which the driver **47** and passenger **45** are located, and may be horizontal along the X-axis of FIG. **1A** for example.

[0244] The coordinate system of x-y-z for the display device **100** may be offset from the coordinate system X-Y-Z for the vehicle **650**. The predetermined axis may differ from the illustrative example. In an alternative embodiment, the predetermined axis may be along a direction tilted with respect to the sides of the display device **100** active area, for example if the driver **47** of FIG. **1A** has a nominal height offset along the Z-axis compared to the passenger **45**.

[0245] Returning to the description of FIG. **3A**, the magnitude of the electric field **370** changes monotonically along a predetermined axis across at least part of the display device **100**. The change may be with monotonic profile **322** of electric field **370** as will be described further hereinbelow. On the left side of the display device **100** voltage $V_{sub.L}$ is provided across the liquid crystal layer **314** to provide corresponding electric field **370** of $E_{sub.L}$ which is higher than the magnitude of the electric field **370** of $E_{sub.C}$ achieved by voltage $V_{sub.C}$ that is applied at the centre of the display device **100**. On the right side of the display device **100**, electric field **370** of $E_{sub.R}$ is provided by voltage $V_{sub.R}$ across the liquid crystal layer **314** which is lower than the magnitude

of the electric field **370** of E.sub.C provided by voltage VC that is applied at the centre of the display device **100**.

[0246] Applied voltage results in an out-of-plane rotation of liquid crystal material **414** in locations through the liquid crystal layer **314** that are separated from the alignment layers **419A**, **419B**.

[0247] In the present embodiment, the in-plane components **417Ap**, **417Bp** of liquid crystal alignment at the alignment layers **419A**, **419B** are constant across the area of the display device **100**, that is the alignment layers **419A**, **419B** are uniformly processed during manufacture, for example by uniform rubbing or uniform photoalignment methods.

[0248] A full description of operation of the layer of liquid crystal layer at each point on the display surface is determined by evaluating the propagation of phase fronts through twisted and tilted liquid crystal layers. However, for purposes of illustration, the operation of the layer **314** may be determined by considering the optical alignment of liquid crystal material **414** in a plane **410** that is halfway between the alignment layers **419A**, **419B**. Such an alignment may represent an average alignment of the liquid crystal material **414** in the direction perpendicular to the layer **314**. Thus considering the liquid crystal material **414** alignment within the layer **314**, components **417p** of the optical axis **417** of the liquid crystal material **414** in the plane of the layer **314** of liquid crystal material **414** have an average direction that is along the predetermined direction. Thus components **417Rp**, **417Cp** and **417Lp** are illustrated as having an average direction that is along the x-axis.

[0249] On the left-hand side of the polar control retarder **300**, the optical axis **417L** has an in-plane component **417Lp** that is pointed towards the left-hand direction and is inclined at an angle $p.sub.L$ to the plane **410**; in the centre of the polar control retarder **300**, the optical axis **417C** has an in-plane component **417Cp** that is pointed towards the left-hand direction and is inclined at an angle $p.sub.C$ to the plane **410**; and on the right side of the polar control retarder **300**, the optical axis **417R** has an in-plane component **417Rp** that is pointed towards the left-hand direction and is inclined at an angle $p.sub.R$ to the plane **410**. The angles $p.sub.L$, $p.sub.C$, $p.sub.R$ are different due to the different applied voltages $V.sub.L$, $V.sub.C$, $V.sub.R$ that provide electric fields $E.sub.L$, $E.sub.C$, $E.sub.R$ respectively wherein $E.sub.L > E.sub.C > E.sub.R$, in this illustrative embodiment.

[0250] In other words, the magnitude of the electric field **370** changes monotonically along a predetermined axis across at least part of the display device **100** so that directions **448** of minimum light transmission of the view angle control arrangement **310** from points of said at least part of the display device **100** are directed towards a common off-axis point **427** in front of the display device **100**.

[0251] The transmission of various rays transmitted through the switchable liquid crystal retarder **301** to the passenger **45** varies as **446R**, **446C**, **446L** and the transmission of the switchable liquid crystal retarder to the driver **47** varies as **448R**, **448C**, **448L**.

[0252] The luminance profile for rays **445**, **447** directed towards passenger **45** and driver **47** will now be described further by illustrating schematically the propagation of polarised light from the output polariser **218** for on-axis and off-axis directions.

[0253] FIG. **3B** is a side view of propagation of output light from the SLM **48** through the optical stack of FIG. **2A** in a privacy mode. Features of the arrangement of FIG. **3B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0254] For illustrative purposes only, in comparison to the arrangement of FIG. **3A** and TABLE 1, the polarisation orientations are illustrated as being in the plane of the paper or out of the plane of the paper. However, in practice the 45° and 135° orientations are used in the present embodiment.

[0255] Light rays from backlight **20** are output unpolarised or partially polarised. Additional polariser **318** transmits polarisation state with electric vector transmission direction **319**.

[0256] Considering light rays to the driver **47**, light ray **447C** that passes from point **470C** to the driver **47** undergoes substantially no phase shift so that the output polarisation state **364C** remains substantially the same as the polarisation state **319**. Such light is absorbed at the input display

polariser **210**, and thus the driver sees low luminance in the direction **447C**. Light rays **447R**, **447L** that pass from points **470R**, **470L** respectively to the driver **47** also undergo substantially no phase shift so that the output polarisation state **364R**, **364L** remain substantially the same as the polarisation state **319**. Such light is absorbed at the input display polariser **210**, and thus the driver sees low luminance in the direction **447R**, **447L**. Note that the angles **457R**, **457C**, **457L** from the normal **199** to the plane of the display device **100** of output of the rays **447R**, **447C**, **447L** vary across the width of the polar control retarder **301**, in correspondence with the field of the view of the display device **100** as seen by the driver **47**. As will be described hereinbelow, the uniformity of reduced luminance to the driver **47** in privacy mode is advantageously increased, achieving increased uniformity of security factor, S.

[0257] Considering light rays to the passenger **45**, light ray **445C** that passes from point **470C** to the passenger **45** undergoes a phase shift so that most generally an elliptical output polarisation state **362C** is incident on the display input polariser, at least some of which is transmitted by the display polariser **210** to the passenger **45**. The passenger sees high luminance modulated with the image data of the SLM **48** in the direction **445C**. Light rays **445R**, **445L** that pass from points **470R**, **470L** respectively to the passenger **45** also undergo a similar phase shift so that the output polarisation state **362R**, **362L** has a similar elliptical polarisation state. Such light is transmitted at the input display polariser **210**, and thus the passenger sees a high luminance in the direction **445R**, **445L** with similar luminance to that in direction **445C**. Note that the angles **455R**, **455L**, **455C** from the normal **199** to the plane of the display device **100** of output of the rays **445R**, **445L**, **445C** vary across the width of the polar control retarder **301**, in correspondence with the field of the view of the display device **100** as seen by the passenger **45**. As will be described hereinbelow, the uniformity of high luminance to the passenger **45** in privacy mode is advantageously increased, achieving increased observed image uniformity.

[0258] FIG. **3C** is a perspective view of a polar control retarder that may be applied in FIG. **2A**, comprising a twisted nematic switchable LC retarder and uniform voltage in the lateral direction across at least one region of the polar control retarder in a privacy mode of operation. Features of the arrangement of FIG. **3C** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0259] In comparison to the arrangement of FIG. **3BA**, a common electric field $E_{\text{sub.L}}$, $E_{\text{sub.C}}$, $E_{\text{sub.R}}$ may be applied perpendicular to the layer **314** of liquid crystal molecules **414** and the magnitude of the electric field is uniform across at least part of the display device **100**.

[0260] The in-plane components **417Lp**, **417Cp**, **417Rp** are the same and directed towards the left-hand direction and is inclined at common angles $p_{\text{sub.L}}$, $p_{\text{sub.C}}$, $p_{\text{sub.R}}$ to the plane **410**. The angles $p_{\text{sub.L}}$, $p_{\text{sub.C}}$, $p_{\text{sub.R}}$ are different due to the common applied voltages $V_{\text{sub.L}}$, $V_{\text{sub.C}}$, $V_{\text{sub.R}}$ respectively. Light rays **448L**, **448C**, **448R** propagating along parallel directions are directed towards the driver **47** and light rays **446L**, **446C**, **446R** propagating along parallel directions are directed towards the passenger **45**.

[0261] The embodiment of FIG. **3C** may be advantageously achieve lower cost and complexity in comparison to the embodiment of FIG. **3A**.

[0262] It may be desirable to provide further control of the common off-axis point **427**.

[0263] FIG. **3D** is a schematic perspective side view of a switchable liquid crystal retarder **301** that may be applied in FIG. **2A**, comprising a twisted nematic switchable LC retarder with alignment layer orientations **417A**, **417B** that vary across the lateral direction, and further comprising a profile **322** of voltage in the lateral direction across at least one region of the switchable liquid crystal retarder **301**; and FIG. **3E** is a graph of in-plane angle **617** against position in the predetermined (lateral) direction and illustrates exemplary profiles **604**, **606**, **608** of the in-plane components **417Ap**, **417Bp** provided by a surface alignment layers **419A**, **419B** of FIG. **3D**. Features of the arrangements of FIGS. **3D-E** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0264] In the alternative embodiment of FIG. 3D, the surface alignment layer **419A** is arranged to provide homogenous alignment in the adjacent liquid crystal material **414** the surface alignment layer **419A** having an in-plane component **417ALp**, **417ACp**, **417ARp** in the plane of the layer **314** of liquid crystal material **414** having an angle **617ALp**, **617ACp**, **617ARp** that changes monotonically along a predetermined axis across at least part of the display device **100**; and the surface alignment layers **419B** is arranged to provide homogenous alignment in the adjacent liquid crystal material **414** the surface alignment layer **419B** having an in-plane component **417BLp**, **417BCp**, **417BRp** in the plane of the layer **314** of liquid crystal material **414** having an angle **617BLp**, **617BCp**, **617BRp** that changes monotonically along a predetermined axis (such as the x-direction in FIG. 3D) across at least part of the display device **100**.

[0265] In other words, the two surface alignment layers **419A**, **419B** are disposed adjacent to the layer **314** of liquid crystal material **414** and on opposite sides thereof wherein the surface alignment layers **419A**, **419B** are arranged to provide alignment in the adjacent liquid crystal material **414** with an in-plane component **417Ap**, **417Bp** that is in the plane of the layer of liquid crystal material, wherein the angle of said in-plane component **417Ap**, **417Bp** of the alignment in the adjacent liquid crystal material **414** changes monotonically along a predetermined axis across at least part of the display device **100**. Further, the components **417ALp**, **417ACp**, **417ARp** vary monotonically across the lateral direction on the alignment layer **419A**, and the components **417BLp**, **417BCp**, **417BRp** vary monotonically across the lateral direction on the alignment layer **419B**, wherein the liquid crystal material **414** twists between the alignment layers **419A**, **419B**.

[0266] The embodiments of FIGS. 3D-E provide for further control of common off-axis point **427**. Such further control is provided by modifying the angle **617** of the respective alignment layer **419A**, **419A** in-plane components **417Ap**, **417Bp** during manufacture. In the embodiments of FIG. 3E, the profile of angle **617** may vary across the predetermined direction as shown by illustrative profiles **604**, **606**, **608** to provide desirable control of the common off-axis point **427**.

[0267] Said further control controls the transmission by the polarisation control retarder **300** with respect to the polar viewing angle across the predetermined direction. For example, considering FIG. 4A hereinbelow, the direction of maximum transmission of the polar control retarder **300** may be different across the predetermined direction, for example the lateral direction.

[0268] In comparison to the embodiment of FIG. 2A, said further control is fixed at manufacture of the alignment layers **419A**, **419B** and control is also provided by modification of the voltage profile **322** provided across the electrodes **320A**, **320B** and across the layer **314** of liquid crystal material **414**.

[0269] The alternative embodiments of FIGS. 3C-D may be provided with each of the embodiments described herein comprising voltage profile **322**. By way of comparison with the embodiment of FIG. 2A, the control provided by the profile **322** may be provided by a smaller voltage gradient across the lateral direction. Advantageously the cost and complexity of the transparent electrodes **320A**, **320B** and the control system **352** may be reduced.

[0270] Control of the common off-axis point **427** by means of alignment layer **419A**, **419B** arrangement is described further in U.S. Pat. No. 11,079,646, which is herein incorporated by reference in its entirety.

[0271] FIGS. 4A-C are graphs illustrating the variation of transmission with polar direction for the polar control retarder **300** of FIG. 2A using the twisted nematic polar control retarder **300** of FIG. 3A and TABLE 1 for different points on the display device **100**.

[0272] FIGS. 4A-C represent the polar variation in relative transmission of the polar control retarder **301** of TABLE 1 for three illustrative drive voltages $V_{sub.R}$, $V_{sub.C}$, $V_{sub.L}$ respectively. Region **435** represents the angular field of the outline of a 12.3" passenger infotainment display device **100** viewed from 750 mm and from a head-on position. Region **437** represents the angular field of the outline of the same display device **100** viewed from the same 750 mm viewing plane and from an off-axis position centred at 30 degrees off-axis in the lateral

direction.

[0273] The angular locations **455R**, **457R**, **455C**, **457C**, **455L**, **457L** of FIG. 3B for zero elevation are marked on respective plots. It can be seen that the transmission of approximately 80%~90% is achieved for all three points **455R**, **455C**, **455L**, and the transmission of approximately 15%~25% is achieved for each of the locations **457R**, **457C**, **457L**. If the driver **47** were to move to higher off-axis angles, then the corresponding transmission would fall; however, the uniformity of low transmission across the display area will be substantially maintained. Advantageously security factor uniformity is increased in comparisons to polar control retarders **301** with no voltage profile **322**.

[0274] In the embodiment of FIG. 3C, a single voltage may be applied, and one of the polar profiles of transmission of FIGS. 4A-C may be provided across at least part of the display device **100**.

[0275] The operation of the polar control retarder in a share mode of operation will now be described.

[0276] FIG. 5A is a perspective view of the polar control retarder **300** of FIG. 2C, comprising a twisted nematic switchable LC retarder **301** and no voltage applied across the liquid crystal layer for a share mode of operation. Features of the arrangement of FIG. 5A not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0277] FIG. 5A is similar in structure to FIG. 3A and FIG. 3C other than no voltage is applied across any of at least one region of the liquid crystal layer **314** for at least part of the display device **100**.

[0278] Thus at the intermediate plane **417L**, the directors **417L**, **417C**, **417R** of the liquid crystal molecules lie uniformly within the plane **410** and a substantially uniform transmission profile is achieved across the polar control retarder.

[0279] FIG. 5B is a side view of propagation of output light from the SLM **48** through the optical stack of FIG. 2A in the share mode. Features of the arrangement of FIG. 5B not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0280] As for FIG. 3B, for illustrative purposes only, in comparison to the arrangement of FIG. 5A and TABLE 1, the polarisation orientations are illustrated as being in the plane of the paper or out of the plane of the paper. However, in practice the 45° and 135° orientations are used in the present embodiment.

[0281] In comparison to the explanation of FIG. 3B, the polarisation state **319** is rotated to output polarisation state **362** across at least a region of the polar control retarder **301**. The light rays **445**, **447** are transmitted by the input display polariser **210** and transmitted to the passenger **45** and driver **47**, modulated with the image data of the SLM **48**.

[0282] FIG. 5C is a schematic graph illustrating the variation of transmission with polar direction for the polar control retarder **300** of FIG. 16A with no voltage V profile **322** across the polar control retarder **300**.

[0283] For each of the subtended display areas, the uniformity of transmission is advantageously high, and high image uniformity of image appearance is achieved.

[0284] An alternative arrangement of additional polariser **318** and display polariser **210** will now be described.

[0285] FIG. 5D is a schematic graph illustrating the variation of transmission with polar direction for a polar control retarder **300** using the twisted nematic polar control retarder **301** with additional polariser **318** and display polariser **210** that have electric vector transmission directions **319**, **211** that are parallel and operating in privacy mode and for the embodiment of TABLE 2; and FIG. 5E is a schematic graph illustrating the variation of transmission with polar direction for a polar control retarder **300** using the twisted nematic polar control retarder **301** with additional polariser

318 and display polariser **210** that have electric vector transmission directions **319**, **211** that are parallel operating in share mode and for the embodiment of TABLE 2 but with 10V uniform drive voltage of the liquid crystal layer **314** compared to the zero volts for FIG. 5C.

TABLE-US-00002

Item	angle	angle	Twist layers	Pretilt	$\Delta n.d$	$V_{sub.C}$	$\phi_{sub.A}$	$\theta_{sub.A}$	$\phi_{sub.B}$	$\theta_{sub.B}$
318	319	45°	314	417Ap	45°	210	211	45°		
90°	Homogeneous	2°	500 nm	1.55 V	417Bp	135°	Homogeneous	2°	210	211

[0286] In privacy mode, the luminance profile for the passenger **45** is extended, advantageously achieving increased image uniformity in the vertical direction for the passenger **45**.

[0287] It would be desirable to reduce the cost of the optical stack to achieve increased viewing freedom of driver **47** for high security factor, S.

[0288] FIG. 6A is a schematic top view illustrating schematically the operation for a driver **47** of a privacy display device **100** comprising the polar control retarder **300** with a voltage profile **322a** across the polar control retarder **300** and viewer tracking system **200** arranged to adjust the voltage profile **322a** to voltage profile **322b** in response to the measured location of the driver **47**; and FIG. 6B is a schematic graph illustrating the variation in voltage V profile **322** applied to the polar control with respect to lateral position across the polar control retarder **300** for two different driver **47** lateral locations and for a display of width L in the predetermined direction (x-axis in this embodiment). Features of the arrangement of FIGS. 6A-B not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0289] In the embodiment of FIGS. 6A-B, the viewer tracking system **200** is arranged to track the location of a viewer **47** movement **540** in the predetermined direction (laterally), wherein the control system **500** is arranged to vary the voltages to the first and second electrode arrangements **320A**, **320B** in response to the location of the viewer **47**.

[0290] Thus the control system **500** is arranged to control the spatial light modulator **48** and to supply voltages to the first and second electrode arrangements **320A**, **320B** for providing the electric field perpendicular to the layer **314** of liquid crystal material **414**, the control system being arranged to vary the voltages to the first and second electrode arrangements **320A**, **320B** for controlling a direction of minimum light transmission of the view angle control arrangement **310**.

[0291] In operation, the voltage profile **322a** with left and right side voltages $V_{sub.La}$, $V_{sub.Ra}$ is provided to the polar control retarder **300** for a first measured viewing position, such that a uniformly low security factor is provided across the display device **100**. After movement **540** of the driver **47** away from the optical axis **199**, the voltage profile **322a** is adjusted by voltage change **348** to maintain the location of the minimum transmission to the driver **47** with voltage profile **322b** with left and right side voltages $V_{sub.Lb}$, $V_{sub.Rb}$.

[0292] For the first voltage profile **322a**, a first point **427a** is provided for points across at least part of the display device **100** which is towards the driver **47**. Similarly for the second voltage profile **322b**, a second displaced point **427b** is provided for points across at least part of the display device **100**. The transmission directed towards the driver **47** may advantageously be minimised in response to the location of the driver across at least part of the display device **100**.

[0293] FIG. 6C is a schematic top view illustrating schematically the operation for a driver of a privacy display comprising the polar control retarder with a voltage profile across the polar control retarder and a head tracking system arranged to adjust the voltage across the polar control retarder in response to the measured location of the driver moving towards the display; and FIG. 6D is a schematic graph illustrating the variation in voltage profile applied to the polar control with respect to lateral position across the polar control retarder for two different driver lateral locations for the driver moving towards the display. Features of the arrangement of FIGS. 6C-D not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0294] In the embodiment of FIGS. 6C-D, the viewer tracking system **200** is arranged to track the

location of a viewer **47** movement **540** in towards the centre of the display, wherein the control system **500** is arranged to vary the voltages to the first and second electrode arrangements **320A**, **320B** in response to the location of the viewer **47**. In operation, the voltage profile **322a** with left and right side voltages $V_{sub.La}$, $V_{sub.Ra}$ is provided to the polar control retarder **300** for a first measured viewing position with a first monotonic profile **322a**, such that a uniformly low security factor is provided across the display device **100** for the driver near point **427a**. After movement **540** of the driver **47** in this illustrative example towards the display device **100** centre, then light rays **447C** to point **427b** are in substantially in the same direction for both viewing positions. In the centre of the display device **100**, the voltage is thus unmodified. However light rays **447R**, **447L** have different angles so that the profile **322b** is modified accordingly with respect to profile **322a** as illustrated in FIG. **6D**.

[0295] The transmission directed towards the driver **47** by view angle control element **310** of display device **100** may advantageously be minimised in response to the location of the driver across at least part of the display device **100**. Uniformity of security factor may be increased over a wider driver **47** headbox.

[0296] It may be desirable to reduce the complexity of the polar control retarder **300**.

[0297] FIG. **6E** is a schematic top view illustrating schematically the operation for a driver **47** of an alternative privacy display device **100** comprising the polar control retarder **300** with a common voltage V across the polar control retarder **300** and a head tracking system arranged to adjust the common voltage V across the polar control retarder **300** in response to the measured location of the driver **47**; and FIG. **6F** is a schematic graph illustrating the variation in common voltage V applied to the polar control for two different driver **47** lateral locations for the arrangement of FIG. **6E**.

[0298] In comparison to the arrangement of FIGS. **6A-B**, the magnitude of the electric field perpendicular to the layer **314** of liquid crystal material is uniform across the entirety of the display device **100**.

[0299] Luminance reduction to a moving driver **47** may be provided by tracking the driver **47** polar location **448** and adjusting the voltage across the liquid crystal layer of the polar control retarder in correspondence. High image security may be provided at the user. At a first measured polar location **448a** of the driver **47**, then profile **322a** with constant voltage V_a is applied and at a second measured polar location **448b** of the driver **47** then profile **322b** with constant voltage V_b is applied.

[0300] The luminance supplied to the driver may be minimised for at least one location on the display device **100**. Reduced luminance is achieved over an increased polar region. Advantageously the cost and complexity of the transmissive electrodes **320A**, **320B** and drive circuit **352** may be reduced.

[0301] An illustrative example of tracking of the location of the driver **47** using controllable transmission profile of polar control retarder **300** will now be described.

[0302] FIG. **7A** is a schematic top view of operation of a privacy display device **100** for a moving driver **47** and a head tracking system arranged to minimise luminance to the driver **47**; and FIG. **7B-D** are graphs illustrating the variation of transmission with polar direction for the polar control retarder **300** of FIG. **2A** using the twisted nematic polar control retarder **300** of FIG. **3** operating with different voltage V profiles **322** across the polar control retarder **300** and the embodiment of TABLE 2.

[0303] As illustrated for FIGS. **7A-D**, luminance reduction to a moving observer may be provided by tracking the driver **47** polar location **448** and adjusting the voltage profile of the polar control retarder in correspondence.

[0304] For the location **47B**, the average voltage across electrodes **320A**, **320B** may be adjusted to provide transmission profile of FIG. **7B** of the view angle control element; for the location **47C**, the average voltage across electrodes **320A**, **320B** may be adjusted to provide transmission profile of FIG. **7C** of the view angle control element; and for the location **47C**, the average voltage across

electrodes **320A**, **320B** may be adjusted to provide transmission profile of FIG. **7C** of the view angle control element.

[0305] Advantageously high image security for multiple driver locations **47** is achieved as will now be described.

[0306] FIG. **8A** is a schematic graph illustrating a simulated polar profile of the security level, S of the arrangement of FIG. **2A** for an ambient illuminance measured in lux that is equal to the head-on display luminance measured in nits and for the polar profile of backlight luminance of FIG. **16A** hereinbelow, and polar profile of polar control retarder transmission of FIG. **7C**, with Fresnel front surface reflectivity of the display device **100**; and FIG. **8B** is a schematic graph illustrating a simulated polar profile of the security level, S of the arrangement of FIG. **2A** for an ambient illuminance measured in lux that is equal to the head-on display luminance measured in nits and for the polar profile of backlight luminance of FIG. **16A** and polar profile of polar control retarder transmission of FIG. **7D**, with Fresnel front surface reflectivity of the display device **100**.

[0307] Region **750** represents the polar region for the passenger **45** to achieve high image visibility with $S < 0.2$. The angular locus of display outline **435** is thus provided in the region of high image visibility and the passenger can easily see the image data from the SLM **48**. Region **752** comprises the polar region over which the image is neither clearly visible or clearly private with $0.2 \leq S < 1$. Region **754** comprises the polar directions over which most image content appears to be private with $1 \leq S < 1.8$ while observers located in region **754** see no images, independent of content.

[0308] In the present embodiments using the view angle control arrangement **310** of FIG. **3A** for example, the polar control retarder **300** is pupillated so that the security factor at the centre of the display is similar to that at the edges of the display, and thus the driver **47** may see a uniformly private image ($S=1$) for a viewing angle of 25° .

[0309] In systems without pupillation then parts of the display region **437** will have $S < 1$ and so parts of the display will appear to be non-private.

[0310] In regions **758** for which $S < 1$ at higher viewing angles for the driver, the image may appear to drop below a threshold of acceptable privacy.

[0311] The variation in location of polar region **754** with acceptable security factor may be provided by control of voltage profile **322**. Thus the arrangement of FIG. **8A** using the transmission profile of FIG. **4B**, may be adjusted by changing the voltage profile **322** by change **348** to achieve the security factor profile of FIG. **8B** as the driver **47** moves closer to the axis, that is closer to the angle α in FIG. **1A**.

[0312] It would be desirable to provide further improvements in image security in the polar region near to the driver **47**. In embodiments hereinbelow such as illustrated in FIGS. **23A-C**, FIG. **24A** and FIG. **25A** increased image security may be provided by using a further additional polariser and a further polar control retarder. Advantageously the embodiment illustrated by the results of FIGS. **8A-B** may be achieved with fewer polar control retarders **300**, achieving reduced cost, thickness, weight and complexity.

[0313] Methods to control a polar control retarder **300** will now be described.

[0314] FIG. **9A** is a flow chart illustrating a method to control an observer-tracked polar control retarder **300**. Features of the arrangement of FIG. **9A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0315] The location of the observer **47** may be determined by means of viewer tracking system **200** and associated image processing. The ambient light level may be measured by means of ambient light sensor **204**. The ambient light level may be used as an input to set the display brightness so that the viewability of the display device **100** may be optimised to a minimum or a maximum as desired. The ambient light system **204** may detect levels impinging on it as a function of input angle. The control system **500** may be set to track the observer **47** by processing sensor data **910**.

[0316] The control system **500** may be operated with both pupillation and tracking control of the

voltage profile **322** across electrodes **320A**, **320B**.

[0317] Alternatively the control system **500** may be operated without pupillation needing to be adjusted in response to driver **47** location. The voltage profile may be adjusted for movement of the driver **47** in the predetermined direction by modifying the average voltage of the profile, for example by means of voltage change **348** as illustrated in FIG. **6B** or FIG. **6D**.

[0318] All or optionally part of the display device **100** may be set or determined **912** to be pupillated by options in the control system **500**. The pupillation may be controlled by an adjustment **914** of voltages on the polar control retarder **300** of the view angle control arrangement **310** as set by system **500** control options.

[0319] In the case where the pupillation is not tracked, for example as illustrated in FIGS. **6E-F** a similar method with appropriately adjusted voltages applied to the polar control retarder **300** of the view angle control arrangement **310** to optimise the display security factor for the driver **47** location is performed as illustrated on the right-hand side of FIG. **9A**.

[0320] The observer **47** location signal from viewer tracking system **200** may be mathematically differentiated with respect to time in order to determine velocity and acceleration of the observer **47** so that any of this data can be used in a predictive tracking control loop tracking the observer to further improve dynamic tracking performance as the observer **47** moves.

[0321] FIG. **9B** is a flow chart illustrating a method for manual control of an adjustable polar control retarder **300**, for example for use with the arrangement of FIG. **1B**. Features of the arrangement of FIG. **9B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0322] The display device **100**, is not tracked by means of a viewer tracking system **200**, the display device **100** may nevertheless be adjusted and controlled by a manual actuator **202** such as dial or buttons so that these signals may be processed **911** and the viewing conditions set or adjusted by the observer **45** or user. The control of the system may be achieved by means of adjustment of the voltages in the polar control retarder **300**, included in the view angle control arrangement **310**. Advantageously cost and complexity may be reduced.

[0323] The operation of a pupillated display device **100** in privacy mode will now be further described.

[0324] FIG. **10A** is a schematic top view illustrating schematically the operation for a passenger **45** of a privacy display device **100** comprising the polar control retarder **300** with voltage **V** profile **322** of FIG. **2A**; and FIG. **10B** is a schematic top view illustrating schematically the operation for a driver **47** of a privacy display device **100** comprising the polar control retarder **300** with voltage **V** profile **322** of FIG. **2A**. Features of the arrangement of FIGS. **10A-B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals, including any potential variations in the features.

[0325] In comparison to the description of FIG. **3B**, the output from the polar control retarder **300** is illustrated as cones **460R**, **460C**, **460L**, where the width of the respective cone **460R**, **460C**, **460L** represents the angular width comprising a limited range of transmission values, for example the cone angles for which transmission is within 20% of a nominal value.

[0326] As illustrated in FIG. **10A**, the light cones **460R**, **460C**, **460L** towards the passenger **45** may be arranged to overlap in an illumination window **325** at a plane **197** at distance **482** from the display device **100** that may be different from the nominal viewing distance **480** of the passenger **45** from the display device **100** along the display normal **199**. The distance **482** may be greater than the distance **480**. Advantageously the variation in uniformity as the passenger **45** moves in a lateral direction may have a natural sensation of roll-off rather than a rapid switching behaviour.

[0327] Similarly, as illustrated in FIG. **10B** the cones **462R**, **462C**, **462L** of low light transmission towards the passenger **45** may be arranged to overlap in an illumination window **327**, and at a greater distance from the display device **100** than the nominal driver **47** locations.

[0328] The illumination windows **325**, **327** are sometimes referred to as pupils, and the output transmission profile from the polar control retarder provided by the voltage variation **322** may be referred to as pupillated, that is the illumination windows **325**, **327** are at a finite distance from the display device **100**.

[0329] By way of comparison with the embodiment of FIGS. **3A-B** the output of a display without the voltage profile **322** will be described.

[0330] FIG. **11A** is a schematic top view illustrating schematically the operation for a passenger **45** and a driver **47** of a privacy display device **100** comprising the polar control retarder **300** with a common voltage **V** in the lateral direction across at least one region of the polar control retarder **300**; and FIG. **11B** is a schematic graph illustrating the variation of transmission across the display device **100** surface with polar direction for a polar control retarder **300** with a common voltage **V** across the polar control retarder **300** of TABLE 1. Features of the arrangement of FIGS. **11A-B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0331] In comparison with the arrangement of FIG. **3B**, the output is not pupillated and each point **470R**, **470C**, **470L** on the display device **100** directs the maximum transmission in a common direction **458** and the minimum transmission in a common direction **459**.

[0332] FIG. **11B** illustrates that the angles **455L**, **455C**, **455R** of FIG. **3B** have different transmissions and thus the uniformity of the image is reduced for the passenger **45**. Similarly, the transmission to the driver varies with the angles **457L**, **457C**, **457R** of FIG. **3B** have different transmissions and thus the uniformity of security factor is reduced for the driver **47**.

[0333] To achieve the pupillation effects described above it would be desirable to achieve a one-dimensional variation in the electro-optic response of the liquid crystal material **414** of the switchable liquid crystal retarder **301**. An illustrative structure that can achieve the desirable characteristics of the display device **100** of FIG. **2A** will now be described.

[0334] FIG. **12A** is a schematic diagram illustrating a front perspective view a transparent electrode **320A** comprising part of a switchable liquid crystal retarder **301** of a switchable display device **100** comprising a transmissive SLM **48**. Features of the arrangement of FIG. **12A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0335] The first electrode arrangement **320A** comprises a continuous electrode extending over the entirety of the layer **314** of liquid crystal material **414**, the continuous electrode having two contacts **321L**, **321R** disposed at opposite ends of the continuous electrode along the predetermined axis and arranged to supply respective voltages that create a voltage profile **322** on the continuous electrode **320A** along the predetermined axis for providing the electric field **370** perpendicular to the layer **314** of liquid crystal material **414**. In alternative embodiments, the electrode **320B** rather than **320A** may comprise the structure of FIG. **12A**. The voltage profile **322** may be a voltage gradient that is monotonic. The voltage profile **322** provides the variation of electric field **370** across the predetermined direction, for example as illustrated in FIG. **3A**.

[0336] In this embodiment, the second electrode arrangement **320B** is arranged to supply a common voltage across the area of the electrode arrangement **320B**. In other words, the second electrode arrangement **320B** comprises a continuous uniform resistance per square electrode extending over the entirety of the layer **314** of liquid crystal material **414**.

[0337] The control system **352** of FIG. **2A** is arranged to supply voltages to the first and second electrode arrangements **320A**, **320B** for providing the electric field **370** perpendicular to the layer **314** of liquid crystal material **414**.

[0338] In the illustrative example of FIG. **3A**, the electric field **370** perpendicular to the layer **314** of liquid crystal is produced by means of x-axis variation in the electrode voltage in the predetermined direction which is the lateral or x-axis direction leading to a variation in the electro optic response of, for example, a liquid crystal material **414** in the layer **314**.

[0339] In the embodiment of FIG. 12A different voltages VL, VR are applied at the left and right ends of the transparent electrode **320A**. This differs from a conventional liquid crystal display where the voltage applied to the transparent electrode **320A** is either applied at a single point or the same voltage VL is applied at multiple points. This is to establish a uniform potential on the transparent electrode **320A**. Note that in the conventional case there is no current flow in the predetermined direction so the potential across the resistance of the transparent electrode **320** remains the same. By way of comparison in the present embodiments current does flow between electrodes **321L**, **321R**.

[0340] In FIG. 12A different voltage contacts **321** are applied at different points specifically VL and VR where $VL > VR$ is illustrated and a voltage profile **322** across (along the x-axis direction of) the transparent electrode **320A** is produced because of its sheet resistance. Typical sheet resistances for transparent conductive coatings are in the range 10 to 1000 ohms per square. The point contact of the voltage VL and VR may provide a non-uniformity of the magnitude of the electric field around the contact **321L**, **321R** and this can cause the voltage profile **322** to deviate from uniformity in the y-axis near to the contacts. This may be particularly undesirable for low aspect ratio (more square) displays.

[0341] It would be desirable to reduce the deviation from uniformity of the voltage profile **322** near to the contacts **321L**, **321R** of VL and VR.

[0342] FIG. 12B is a schematic diagram illustrating in a front perspective view a transparent electrode **320A** with additional bus bars comprising the contacts **321L** and **321R**. Features of the arrangement of FIG. 12B not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0343] The two contacts **321L**, **321R** comprise conductive bars extending perpendicular to the predetermined axis.

[0344] The bus bars contacts **321L**, **321R** comprise material with higher conductivity than that of the transparent electrode **320A** and may for example comprise metal or a stack of thin film metals such that the voltage VL is effectively applied with a low impedance line contact to the transparent electrode **320A** rather than a point contact. The voltage profile **322** in the y-axis direction is therefore more uniform compared to that of FIG. 12A and the display device **100** can advantageously provide reduced visibility of edge effects from non-uniform electric fields **370** in the y-axis direction.

[0345] The voltage profile **322** could be reversed by making $VR > VL$. The voltage profile **322** could alternatively be produced in the y direction, if so desired, by altering the location of voltage contacts **321L**, **321R** (not illustrated).

[0346] FIG. 12C is a schematic diagram illustrating in front perspective an exemplary method of implementing voltages VL and VR. VL and VR are applied to transparent electrode **320A** by means of driver **330**, **331** which may be an integrated circuit comprising an op-amp. Features of the arrangement of FIG. 12C not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0347] VL may be an alternating potential such as a square or sine wave. The output of drive circuit **330** is scaled down, illustrated schematically by variable resistor **334** and that potential is then buffered by driver **331** to create alternating voltage VR which is synchronised with VL. The voltage scaling may be achieved by other known circuits such that the scaling may be under electronic control by a control system in response to for example the location of an observer. The lateral voltage profile **322s** provided by this circuit creates electric fields in the predetermined direction with magnitudes that are typically much lower than the magnitude of the electric field **370** perpendicular to the switchable liquid crystal retarder **301** i.e. in the z-axis direction.

[0348] FIG. 12A further illustrates Vref that is a reference potential, which may be the potential

applied to transparent electrode **320B** (not shown) for example.

[0349] FIG. **12D** is a schematic diagram illustrating in front perspective an alternative embodiment of a control system **352** for providing voltages VL and VR. Features of the arrangement of FIG. **12D** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0350] VL is applied to transparent electrode **320A** by means of driver **330** which may be an integrated circuit such as an op-amp.

[0351] Bus bar contact **321R** is illustrated schematically connected to tuneable variable resistor **336** and to voltage VG. A current path is created from VL to VG and the relative voltage drop to VR across the sheet resistance of transparent electrode **320A** may be adjusted by means of tuneable resistor **336**. The tuneable resistor **336** may be replaced by an electronically controlled variable voltage that enable the degree of pupillation to be controlled, for example in response to the location of the observer. Voltage VL may be an alternating e.g. square wave voltage in order to meet the DC balance conditions across the liquid crystal layer **314**. Voltage VG may be a fixed potential such as ground or for example an alternating voltage to contribute to the DC balance design of the switchable liquid crystal retarder **301**. VG may also be the potential applied to transparent electrode **320B**.

[0352] FIG. **13A** is a schematic diagram illustrating in front perspective an exemplary assembly of switchable liquid crystal retarder **301** comprising opposing transparent electrodes **320A** and **320B**. Features of the arrangement of FIG. **13A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0353] In comparison with FIG. **12A**, voltages may be applied to multiple contacts **321L**, **321R** on transparent electrode **320B** to create voltage profile **322** and transparent electrode **320A** may have a uniform voltage.

[0354] FIG. **13A** further shows that contacts **321L**, **321R** may be provided at more than one point on at least one side of the electrode arrangement **320A**. Advantageously non-uniformities of electric field profile **370** may be reduced.

[0355] FIG. **13B** is a schematic diagram illustrating in front perspective a further exemplary assembly of switchable liquid crystal retarder **301** comprising opposing transparent electrodes **320A** and **320B**. Features of the arrangement of FIG. **13B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0356] In comparison with reference to FIG. **13A**, bus bars contacts **321L** and **321R** are used, as described with reference to FIG. **12B**, to create a lateral voltage profile **322** on transparent electrode **320B**. The voltage profile **322** may be present on the front or the back of the switchable liquid crystal retarder **301**. The lateral voltage profile **322** creates a difference in the magnitude of the electric field **370** in the perpendicular (z-axis direction) over the layer **314** of liquid crystal material **414**.

[0357] FIG. **13C** is a schematic diagram illustrating in front perspective a further exemplary assembly of switchable liquid crystal retarder **301** comprising opposing transparent electrodes **320A** and **320B**. Features of the arrangement of FIG. **13C** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0358] In this case a voltage profile **322** is produced on both transparent electrode **320A** and **320B** by means of bus bar contacts **321LA**, **321RA** and **321LB**, **321RB** respectively. The voltages VLA, VLB, VRA, VRB applied may be adjusted to give in the required predetermined direction voltage profile **322** to produce the desirable electric field **370** profile in layer **314** of liquid crystal **414** and difference in electro optic response of the liquid crystal in the z-axis of the switchable liquid crystal retarder **301**.

[0359] In the embodiments of FIG. 12A to FIG. 13C the x-axis voltage profile 322 created depends on the applied voltages VL and VR and the resistance uniformity of the respective transparent electrode 320A or 320B. It is common for deposited transparent electrodes to have a uniform resistance per area and therefore the voltage profile 322 across the transparent electrode 320A from bus bar contacts 321L to 321R is essentially linear. However, the electro optic response of the liquid crystal contained in the switchable liquid crystal retarder 301 may require a different response function to achieve the desired optical effect. It would be desirable to be able to produce a prescribed variation in the profile across one or more of the transparent electrodes.

[0360] FIG. 13D is a schematic diagram illustrating different profiles of voltage with lateral position across the display device 100. Features of the arrangement of FIG. 13D not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0361] Voltage profile 322a illustrates a linear voltage profile 322, that is a uniform gradient, between the left and right ends of the electrode layer 320A. However, it may be desirable to produce a non-linear voltage profile 322b between the left and right ends, for example a curve. In this way the voltage VC2 at the centre of the electrode may differ from the linear gradient case voltage VC1. This is so that the electro optic response of the liquid crystal layer 314 varies spatially as desired to provide the desired pupillation of the display. The gradient 322a can be implemented by varying the resistance per square of the electrode material for example.

[0362] The curve 322b may be provided to achieve increased accuracy of directing minimum transmission of light rays to point 427 across the at least part of the display device 100. Advantageously increased uniformity may be provided.

[0363] It would be desirable to be able to implement curve 322b without control during manufacture of the variation of the sheet resistance per square of the area of the electrode material.

[0364] FIG. 13E is a schematic diagram illustrating in front perspective a further exemplary assembly of switchable liquid crystal retarder 301 comprising opposing transparent electrodes 320A and 320B; and FIG. 13F is a schematic graph illustrating an approximation to desired voltage profile 322 curve 322a produced by stepwise approximation using wires 321a-n. Features of the arrangement of FIGS. 13E-F not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0365] The first electrode arrangement 320A comprises plural electrodes 323a-n provided along the predetermined axis, the plural electrodes 323a-n being arranged to supply different voltages that change monotonically along the predetermined axis, for example as illustrated by profile 323a-n of FIG. 13F.

[0366] In this alternative embodiment, the transparent electrode 320A is segmented into a number n of y-axis fingers, each of which is connected by respective wire 321a-n to drive circuit 326, which may be an integrated circuit addressed from a control system (not shown). Drive circuit 326 is arranged to apply a different voltage to each of the respective wires 321a-n so that in principle any desired voltage variation with lateral (x-axis) direction may be approximated, including non-monotonic profiles. Advantageously complexity of the electrode resistance per square variation of 320A is reduced.

[0367] Referring to FIG. 13F, “n” may be any reasonable integer number, for example 256.

[0368] By controlling the voltages on wires 321a-n, areas of the switchable liquid crystal retarder 301 may achieve pupillation and other areas may achieve a different pupillation or no pupillation. This effect can be used to control the viewing of the parts of the display device 100 from different directions. Segmenting plural electrodes 323a-n also provides different parts of the display device 100 to operate to produce different pupillation. In this case the change in voltage within each section is monotonic, but the monotonicity may be broken across the boundary between respective parts.

[0369] It may be desirable to provide further control of profile **322**.

[0370] FIG. **13G** is a schematic diagram illustrating in a perspective front view the opposing electrodes **320A**, **320B** of a switchable liquid crystal retarder **301** comprising corner bus bar contacts **321TL**, **321BL**, **321TR**, **321BR**. Features of the arrangement of FIG. **13G** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0371] In the alternative embodiment of FIG. **13G**, the bus bar contacts **321TL**, **321BL**, **321TR**, **321BR** are provided with shapes and/or locations to provide improved control of profile **322**. Separate drive circuits **330TL**, **330BL**, **330TR**, **330BR** respectively are provided to achieve variations such as shown by illustrative profiles **322TBL**, **322TBR** that are in addition to the lateral profile **322LR**.

[0372] The sheet resistance of the resistive conductive coating (for example ITO) of the electrode **320A** may provide a lower resistance at each of the corners than in the centre of electrode **320A**, so that when viewed from the front, the corners and centre of electrode **320A** may therefore experience a slightly different voltage.

[0373] In operation, the profiles **322TBL**, **322TBR** and **322LR** may be arranged to provide modification of the luminance directed into a common pupil **325** from across the display area such as illustrated in FIG. **6A**. Advantageously uniformity may be increased.

[0374] In alternative embodiments (not illustrated) a similar arrangement may be implemented on electrode **320B**. When the display has a high rectangular aspect ratio, the length of the bus bars **321** may be longer in the x direction, i.e., $\Delta x > \Delta y$ to retain a higher voltage along the edge near the corners of the long side of the rectangle. Advantageously visual uniformity may be improved.

[0375] It would be desirable to further increase the uniformity of the display device **100**.

[0376] FIG. **14A** is a schematic top view illustrating schematically the operation for a passenger **45** of a privacy display device **100** comprising a backlight **20** of FIG. **2A** comprising a first illuminated waveguide **1A**, pupillated turning component **50** and polar control retarder **300**. Features of the arrangement of FIG. **14A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0377] During operation in privacy mode, light rays **447** from light source **15A** are guided through waveguide **1A**, and are directed onto light turning film component **50**, being directed towards the passenger **45**.

[0378] As will be described further hereinbelow with respect to FIGS. **26A-C**, light turning film component **50** may be pupillated, that is arranged to provide light cones **466R**, **466C**, **466L** that vary in nominal pointing direction across the display device **100**. The transmission profiles **460R**, **460C**, **460L** may be arranged in alignment with the profiles **466R**, **466C**, **466L** from both the backlight **20** illumination and for the transmission profiles and pupil **325** may be provided. Advantageously uniformity of luminance provided to the passenger **45** may be increased.

[0379] FIG. **14B** is a schematic top view illustrating schematically the operation for a driver **47** of a privacy display device **100** comprising a backlight **20** of FIG. **2A** comprising a second waveguide **1B** comprising a second luminance profile. Features of the arrangement of FIG. **14B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0380] During operation in privacy mode, stray light rays **451** from light source **15A** are guided through waveguide **1A**, and are directed onto the waveguide **1B**. Such light rays **451** are then output onto the light turning film and directed towards the driver **47** into light cones **468R**, **468C**, **468L** that are pupillated towards driver **47** illumination window (optical pupil) **327** such that the uniformity of stray light is increased.

[0381] Light cones **464R**, **464C**, **464L** similarly provide reduction of the luminance of the light cones **468R**, **468C**, **468L** that is uniform across the display device **100**. The reduction of luminance

can be arranged to minimise the visibility of stray light **451**, advantageously achieving increased uniformity of security factor.

[0382] An alternative arrangement for achieving improved uniformity using a curved display device **100** will now be described.

[0383] FIG. **15A** is a perspective view of a user **45** in front of a curved display device **100** comprising a curved polar control retarder **300** comprising a liquid crystal retarder **301** as described elsewhere herein. Features of the arrangement of FIG. **15A** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0384] In the alternative of FIG. **15A**, the display device **100** is curved about the y-axis and has a centre of curvature on the same side of the display device **100** as the user **45**. At least one of the backlight **20** and polar control retarder **300** may be curved. In the illustrative example of FIG. **15A** both the backlight **20** and polar control retarder **300** are curved.

[0385] By way of comparison with the arrangement of FIG. **15A**, if the polar control retarder **300** and backlight were flat then the respective pupillation properties would be arranged to provide a viewing window **325F** at a window plane distance **197F**. In the present embodiment, the curvature of the display device **100** provides a window plane **325C** at a shorter window plane distance **197C**.

[0386] The voltage profile **322** across the polar control retarder **301** may be modified in correspondence to the curvature of the display.

[0387] The variation of voltage profile **322** across the polar control retarder **301** may be reduced to achieve a desirable window **325C**. The optical transmission profiles of the polar control retarder **301** provided across the display device **100** may have a smaller variation than for the corresponding flat polar control retarder and advantageously increased uniformity may be achieved.

[0388] In alternative embodiments, the curvature may be non-uniform to provide a shaped display to match the profile of the surface of a vehicle, with the voltage profile **322** adjusted accordingly to provide common optical windows **325**, **327** for points across the display device **100**.

Advantageously high image uniformity and security factor uniformity may be provided for displays with non-uniform surface profiles.

[0389] It would be desirable to provide multiple displays that are adjacent to each other. Arrangements of alignment directions for tiled displays will now be described.

[0390] FIG. **15B** is a schematic diagram illustrating a schematic top view of a pair of co-planar tiled display devices **100A**, **100B**; and FIG. **15C** is a schematic diagram illustrating a schematic top view of a pair of tilted planar tiled display devices **100A**, **100B**. Features of the arrangements of FIGS. **15B-C** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0391] In the alternative embodiments of FIGS. **15B-C**, display devices **100A**, **100B** have respective voltage profiles **322A**, **322B**, wherein the profiles **322A**, **322B** are different to desirably achieve common points **427A**, **427B** at or near the driver **47** wherein the transmission of the polar control retarder **300** is minimised. Advantageously uniformity of security factor **S** to the driver and uniformity of image to the passenger **45** may be achieved.

[0392] Common points **427A**, **427B** may be aligned as in the illustrative embodiment of FIG. **15B**. Advantageously the lowest luminance is achieved for a driver **47** at said common location **427A**, **427B**.

[0393] In alternative embodiments the common points **427A**, **427B** may be offset as illustrated in FIG. **15C**. For the first display device **100A**, the visibility of reflected light rays **606A** from ambient light source **604** to the driver **47** may be lower than the visibility of reflected light rays **606B** for the second display device **100B**. For a given luminance reduction at the common points **427A**, **427B**, the security factor of display device **100B** would be higher than for display device **100A**. It would be desirable to provide a uniform security factor for various locations of the driver **47** while the reflectivity varies.

[0394] Driver **47** at point **427A** sees a low luminance from display device **100A** but lower frontal reflections from rays **606A**. By comparison the driver sees higher front reflections from rays **606B** from display device **100B** but higher luminance due to the offset from point **427B**. The offset of common points **427A**, **427B** advantageously achieves a wider viewing region of driver **47** for which the security factor is above a desirable threshold.

[0395] As is apparent from FIGS. **15B-C**, a passenger **45** may no longer be arranged on-axis near to the centre of each display device **100A**, **100B**. Instead, the user may, approximately, be positioned on-axis with an edge of each display device **100A**, **100B**.

[0396] Further in an alternative embodiment the control system **500** may be provided to control the location of the points **427A**, **427B** in response to ambient illumination **604** measured by ambient light sensor **204** and driver **47** location measured by observer position location viewer tracking system **200** to achieve optimised security factor for a measured driver **47** location and illumination conditions.

[0397] In other embodiments the number of parts **101** may be increased to advantageously achieve improved visual performance for privacy and share mode operation.

[0398] In alternative embodiments, at least one of the display devices **100** may be curved as illustrated in FIG. **15A**.

[0399] FIG. **15D** is a schematic diagram illustrating a schematic top view of a segmented curved display device comprising offset common points **427A**, **427B**. Features of the arrangement of FIG. **15D** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0400] In comparison to the arrangement of FIG. **15C**, a single display device **100** may be provided, with part **101A** and part **101B**. Profiles **332A**, **332B** may be provided that are different on the parts **101A**, **101B** respectively to provide different common points **427A**, **427B**. In a similar manner to FIG. **15C**, the security factor across the display may be provided with increased uniformity.

[0401] FIG. **15E** is a schematic diagram illustrating a schematic top view of a segmented curved display device **100** comprising different polar control retarder **300** drive voltages. Features of the arrangement of FIG. **15E** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0402] In comparison with the arrangement of FIG. **15D**, the parts **101A**, **101B** are provided with different drive voltages, but no profile of voltage is provided across the respective parts **101A**, **110B**. The output light rays with minimum luminance **447A**, **447B** may be pupillated at least to an extent by the curvature of the display device **100** as described in FIG. **15A** above.

[0403] The direction of light rays **447A**, **447B** from respective sides of the display parts **101A**, **101B** may be different. The display device **100A** may have an electrode arrangement **320** that is advantageously less complicated than for FIG. **15D**. The directions of light rays **447A**, **447B** may be modified by control of the voltages $V_{sub.A}$, $V_{sub.B}$ across the liquid crystal layer **314** of the respective parts **101A**, **101B** that are different. The voltages $V_{sub.A}$, $V_{sub.B}$ may be arranged to achieve improved security factor for moving driver **47**, and may be in response to measured driver **47** location and/or in response to ambient illumination from ambient light source **604** in a similar manner to that described in FIG. **15D**. Advantageously the security factor uniformity may be improved and cost may be reduced.

[0404] It may be desirable to provide displays with alternative arrangements of curvature.

[0405] FIG. **15F** is a schematic diagram illustrating a schematic top view of a segmented curved display device **100** comprising a planar part **101A** and a curved part **101B**. Features of the arrangement of FIG. **15F** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0406] In the alternative embodiment of FIG. 15F, the display device **100** may be arranged to provide an aesthetically desirable profile across the dashboard of a vehicle **600** for example.

[0407] The profiles **322A**, **322B** may be modified to provide the desirable common points **427A**, **427B** as described hereinabove. Planar part **101A** may be arranged with a profile **322A** that is different to the profile **322B** to achieve points **427A**, **427B** that in the illustrative example of FIG. 15F are coincident. Advantageously a uniform display uniformity may be achieved for the passenger **47**, and the driver **45** may see a display **100** with no driver distraction.

[0408] In alternative embodiments, not shown, the parts **101A**, **101B** may be provided by separate displays **100A**, **100B**. Advantageously yield of manufacture may be increased and replacement cost reduced.

[0409] In alternative embodiments, the part **101A** may be arranged to provide images that are arranged to be seen by the driver, and the profile **322A** may provide a uniform voltage profile across the liquid crystal retarder **301**. Advantageously cost and complexity may be reduced.

[0410] The operation of the backlight **20** of FIG. 2A will now be further described.

[0411] FIG. 16A is a schematic graph illustrating the simulated polar variation of luminance output for an illustrative backlight **20** of FIG. 2A and FIG. 26A primarily operating to direct light to the passenger **45**, that is light source **15A** is illuminated and light source **15B** is not illuminated.

[0412] Advantageously most light is directed towards the passenger **45** in direction **445** and high suppression of luminance is achieved in the driver **47** in direction **447**. High power efficiency for passenger **45** illumination is achieved. However, such an illumination profile is not sufficient to achieve desirable security factor, S as will be described further hereinbelow.

[0413] FIG. 16B is a schematic graph illustrating the polar variation of luminance output for an illustrative alternative backlight **20** of FIG. 2A and FIG. 26A primarily operating to direct light to the driver **47**, that is light source **15B** is illuminated and light source **15A** is not illuminated. Advantageously most light is directed towards the driver **47** in direction **447** and low luminance is achieved in the passenger **45** direction. High power efficiency for driver **47** illumination is achieved in a mode of operation in which the light source **15B** is illuminated.

[0414] FIG. 17A is a schematic graph illustrating the polar variation of luminance output for an illustrative backlight **20** of FIG. 2A and FIG. 26A operating to direct light to the passenger **45** and to the driver **47**, that is both light sources **15A**, **15B** are illuminated; and FIG. 17B is a schematic graph illustrating the variation of relative luminance output at zero elevation for the alternative backlight profile of FIG. 17A operating to direct light to the passenger **45** and driver **47** in a share mode of operation. Such a profile may be provided by a backlight **20** as will be described with reference to FIG. 26A hereinbelow for example.

[0415] The backlight **20** in a second mode of operation has a second luminance distribution having an output luminance profile having first and second maxima at polar locations **466**, **468** in luminance at first and second different polar locations **445**, **447** with a minimum **524** in luminance therebetween. The luminance of the minimum **524** is desirably at most 25% of the luminance of the first and second maxima.

[0416] Advantageously a display device **100** that may be observed by both driver **47** and passenger **45** may be provided.

[0417] Other arrangements of display will now be described.

[0418] FIG. 18 is a schematic top view of a centre stack display for an automotive vehicle **650**. Features of the arrangement of FIG. 18 not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0419] In the alternative embodiment of FIG. 18, display device **100** comprises a right-hand part **101P** that is a switchable privacy display arranged to provide privacy and share modes; and a left-hand part **101D** that is a fixed share mode display. Other arrangements of mixed function display may be provided, for example to provide switchable privacy functions to each of driver **47** and

passenger **45** on respective sides of the display.

[0420] In the alternative embodiment of FIG. **18**, the left part **101D** of the display is provided such that the magnitude of the electric field across the layer **314** of liquid crystal material is uniform across the area of the polar control retarder **300**. Such a uniform region could for example be provided with no transparent electrode, or with a segmented electrode that is separately driven for the part **101D** and **101P**. Both driver **47** and passenger **45** may see a high luminance image provided by high transmission from the polar control retarder **300** over a wide polar range, such as illustrated in FIG. **5C** for example.

[0421] By comparison the region of the polar control retarder **300** arranged for the right-hand part **101P** of the display may be provided with a voltage profile **322** such that a pupillated low transmission profile is directed to the driver to achieve high uniformity of security factor for the right-hand part **101P** of the display device **100** while achieving high image uniformity for the passenger **45** in the manner described hereinabove.

[0422] Various alternative embodiments of display device **100** structures will now be described.

[0423] It may be desirable to use an SLM **48** with polariser **210**, **218** electric vector transmission directions **211**, **219** that are not at 45 degrees to the edge of the active area of the display device **100**.

[0424] FIG. **19A** is a schematic side perspective view of a display device **100** providing uniformity for a display user wherein the polariser transmission directions of the display polarisers are different to those of FIG. **2A**; and FIG. **19B** is a front view of the stack of layers of the display device **100** of FIG. **19A**. Features of the arrangement of FIGS. **19A-B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0425] The at least one polar control retarder **300** further comprises passive retarders **332A**, **332B** that are half wave plates that may be an A-plate with a retardance for light of a wavelength of 550 nm in a range of 250 nm to 300 nm.

[0426] Backlight **20** typically provides partially polarised light **420** incident onto the additional polariser **318**. In the present embodiments, desirably the electric vector transmission direction **319** of the additional polariser **318** is aligned to the linear polarisation transmission direction of the backlight **20**. It would be desirable to provide electric vector transmission direction **319** of additional polariser **318** to be at 0 degrees.

[0427] In the alternative embodiment of FIGS. **19A-B**, half wave plate retarder **332A** has an optical axis direction **333A** that is inclined at 67.5 degrees to the easterly direction, so that the polarisation state of light incident onto the liquid crystal retarder **301** is 135 degrees. Advantageously transmission efficiency of light from the backlight **20** is improved.

[0428] In the embodiment of FIGS. **19A-B**, an SLM **48** input polariser **210** with horizontal transmission direction **211** and vertical output polariser **218** transmission direction **219** is provided. Such polariser arrangements are typically provided for fringe field switching LCDs and vertically aligned nematic LCDs. Advantageously wide angle contrast of images may be improved compared to the TN LCD of FIG. **2A**. Further the output polarisation state **219** may be aligned to the transmission of polarised sunglasses so that advantageously display brightness is improved in displays where the display user is wearing sunglasses.

[0429] In the embodiment of FIG. **19A**, a further passive retarder **332B** has an optical axis direction **333B** that is inclined at 22.5 degrees so that for light rays directed to the point **427**, the output polarisation state from the polar control retarder **300** is aligned to be absorbed by the input polariser **210** as described elsewhere herein.

[0430] In alternative embodiments one of the retarders **332A**, **332B** may be omitted and the additional polariser **318** and display polariser **210**, **218** aligned accordingly with respect to the directions **417Ap**, **417Bp** of the polar control retarder **300**. Advantageously performance may be improved and cost reduced.

[0431] In alternative embodiments, retarders **332A**, **332B** may comprise multi-layer retarders such as Pancharatnum retarders. Advantageously undesirable colouration of transmitted light may be reduced.

[0432] Alternatives to the arrangement of TABLE 1 will now be described.

[0433] FIGS. **20A-F** are graphs illustrating the variation of transmission with polar direction for various arrangements of liquid crystal polar control retarder, display polarisers and additional polarisers.

[0434] More generally, the switchable liquid crystal retarder **301** may have a retardance for light of a wavelength of 550 nm in a range from 300 nm to 1500 nm, preferably in a range from 400 nm to 1200 nm. The twist may be in a range from 60° to 120°, and preferably in a range from 70° to 90°.

[0435] TABLE 3 and FIGS. **20A-B** illustrate an embodiment with 80° twist. In comparison to the embodiment of FIG. **4B**, the amount of light directed to the side passenger window may be reduced. In comparison to the embodiment of FIG. **5C**, share mode performance may be enhanced, with reduced colouration.

TABLE-US-00003 TABLE 3 Active LC retarder 301 In-plane In-plane V.sub.C rotation rotation Alignment (representative Item angle angle Twist layers Pretilt Δn.d profile figure) 318 319, φ.sub.A 130° 314 417Ap, θ.sub.A 130° 80° Homogeneous 2° 650 nm 0 V (FIG. 20A) 417Bp, θ.sub.B 50° Homogeneous 2° 1.6 V (FIG. 20B) 210 211, φ.sub.B 50°

[0436] TABLE 3 and FIGS. **20C-D** illustrate an embodiment with 70° twist. In comparison to the embodiment of TABLE 3, the size of the high transmission viewing zone for the passenger **45** is increased, and on-axis luminance is increased for passenger **45** near to the optical axis **199** of the display. Further if retarders **332A**, **332B** are provided as described with respect to FIGS. **19A-B** then the angle of adjustment of polarisation to match desirable backlight **20** and SLM **48** alignment directions is advantageously reduced, reducing chromaticity and off-axis degradations from said retarders **332A**, **332B**.

TABLE-US-00004 TABLE 4 Active LC retarder 301 In-plane In-plane V.sub.C rotation rotation Alignment (representative Item angle angle Twist layers Pretilt Δn.d profile figure) 318 319, φ.sub.A 125° 314 417Ap, θ.sub.A 125° 70° Homogeneous 2° 1000 nm 0 V (FIG. 20C) 417Bp, θ.sub.B 55° Homogeneous 2° 1.55 V (FIG. 20D) 210 211, φ.sub.B 55°

[0437] TABLE 5 and FIGS. **20E-F** illustrate an embodiment with 80° twist and additional polariser **318** that is crossed to display polariser **210**. In comparison to the embodiment of TABLE 3, the alignment of polarisers may be conveniently achieved without the cost and complexity of retarders **332A**, **332B** such as illustrated in FIGS. **19A-B**.

TABLE-US-00005 TABLE 5 Active LC retarder 301 In-plane In-plane V.sub.C rotation rotation Alignment (representative Item angle angle Twist layers Pretilt Δn.d profile figure) 318 319, φ.sub.A 135° 314 417Ap, θ.sub.A 130° 80° Homogeneous 2° 650 nm 0 V (FIG. 20E) 417Bp, θ.sub.B 50° Homogeneous 2° 1.55 V (FIG. 20F) 210 211, φ.sub.B 50°

[0438] It may be desirable to provide privacy operation for an emissive display.

[0439] FIG. **21** is a schematic side perspective view of a display device **100** comprising an emissive SLM **48** and a polar control retarder **300** and additional polariser arranged to receive light from the SLM **48**. Features of the arrangement of FIG. **21** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0440] In the alternative embodiment of FIG. **21**, the SLM **48** comprises an emissive SLM **48** arranged to emit the spatially modulated light. Pixels **220**, **222**, **224** may be provided by emissive elements such as OLED, micro-LED or other known emitting elements. Advantageously thickness may be reduced in comparison to the embodiment of FIG. **1A**. Said display polariser **210** is an output display polariser **218** arranged on the output side of the SLM **48**.

[0441] Advantageously thickness is reduced in comparison to the embodiment of FIG. **2A**.

Emissive displays also typically advantageously exhibit high image contrast for the passenger **45** in

comparison to conventional LCDs.

[0442] In the embodiment of FIG. 21, there is not a reflective polariser arranged between the output display polariser **218** and the at least one polar control retarder **300**. In liquid crystal modes that are asymmetric in the predetermined direction (such as shown in FIG. 4B), reflectivity of light from a reflective polariser (as will be described hereinbelow with respect to FIG. 23C and FIG. 25A) is reduced because of high input transmission of light from an ambient light source (such as illustrated by source **604** in FIG. 2A) but low reflected luminance. Such reflective polarisers as will be described thus have reduced efficacy for asymmetric modes.

[0443] FIG. 22 is a schematic side perspective view of an optical component comprising a polar control retarder **300** comprising electrodes arranged to provide a variable voltage *V* profile **322** in the lateral direction across at least one region of the polar control retarder **300**. Features of the arrangement of FIG. 22 not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0444] It may be desirable to provide reduction of luminance to the driver over an increased polar range and to a lower light level.

[0445] FIG. 23A is a schematic side perspective view of a display device **100** providing uniformity for a display user **47** comprising first and second polar control retarders **300A**, **300B** with further additional polariser **318B** arranged on the input side of the additional polariser **318**. Features of the arrangement of FIG. 23A not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0446] In comparison to the embodiment of FIGS. 2A-C, in the alternative embodiment of FIG. 23A, the display device **100** further comprises: a further additional polariser **318B** arranged on the same side of the SLM **48** as the additional polariser **318A** outside the additional polariser **318A**, the further additional polariser **318B** being a linear polariser; and at least one further polar control retarder **300B** arranged between the additional polariser **318A** and the further additional polariser **318B**.

[0447] The at least one further polar control retarder **300B** comprises a switchable liquid crystal retarder **301B** comprising a layer **314B** of liquid crystal material **414B**, and the display device **100** further comprises first and second electrode arrangements **320BA**, **320BB** disposed on opposite sides of the layer **314B** of liquid crystal material **414**, wherein first and second electrode arrangements **320BA**, **320BB** are arranged to provide an electric field **370** perpendicular to the layer **314B** of liquid crystal material **414**, wherein the magnitude of the electric field **370** changes monotonically along a predetermined axis across at least part of display device **100**.

[0448] The retardance, twists, and or pre-tilts of the liquid crystal layer **314A**, **314B** may be different for the polar control retarder **300A**, **300B** respectively. The profiles **322A**, **322B** may be different and may provide common points **427** or different points **427** at which transmission uniformity is greatest as described hereinbefore. Additional passive retarders **332AA**, **332AB**, **332BA**, **332BB** (not illustrated) may be provided to achieve desirable polarisation rotations as described elsewhere herein, for example with reference to FIGS. 19A-B.

[0449] Advantageously different polar regions may have reduced transmission, increasing polar region of driver **47** location for high image security.

[0450] The embodiment of FIG. 23A has the SLM **48** arranged between the view angle control arrangement **310** and the output side of the display device **100**. Advantageously reduced visibility of frontal reflections from additional layers may be achieved.

[0451] FIG. 23B is a schematic side perspective view of a display device **100** providing uniformity for a display user comprising first and second polar control retarders **300A**, **300B** with further additional polariser **318B** and second polar control retarder **300B** arranged on the output side of the display output polariser **218**. Features of the arrangement of FIG. 23B not discussed in further

detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0452] In comparison to the embodiment of FIG. 23A, in the alternative embodiment of FIG. 23B, the display device **100** comprises: an output display polariser **218** arranged on the output side of the SLM **48**; a further additional polariser **318B** arranged on the output side of the SLM **48**; and at least one further polar control retarder **300B** arranged between the further additional polariser **318B** and the output display polariser **218**.

[0453] The at least one further polar control retarder **300B** comprises a switchable liquid crystal retarder **301B** comprising a layer **314B** of liquid crystal material **414B**, and the display device **100** further comprises first and second electrode arrangements **320BA**, **320BB** disposed on opposite sides of the layer **314B** of liquid crystal material **414**, wherein first and second electrode arrangements **320BA**, **320BB** are arranged to provide an electric field **370** perpendicular to the layer **314B** of liquid crystal material **414**, wherein the magnitude of the electric field **370** changes monotonically along a predetermined axis across at least part of display device **100**.

[0454] The retardance, twists, and or pre-tilts of the liquid crystal layer **314A**, **314B** may be different for the polar control retarder **300A**, **300B** respectively. The profiles **322A**, **322B** may be different and may provide common points **427** or different points **427** at which transmission uniformity is greatest as described hereinbefore. Additional passive retarders **332AA**, **332AB**, **332BA**, **332BB** (not illustrated) may be provided to achieve desirable polarisation rotations as described elsewhere herein, for example with reference to FIGS. 19A-B.

[0455] Advantageously different polar regions may have reduced transmission, increasing polar region of driver **47** location for high image security.

[0456] The display device **100** may further comprise a reflective polariser **302** arranged between the output polariser **218** and the at least one polar control retarder **300**, the reflective polariser being a linear polariser arranged to pass the same linearly polarised polarisation component as the output polariser **218**. The polarisers **218**, **318B** may have electric vector transmission directions that are parallel and the polar control retarder **300B** may be arranged to provide a more rotationally symmetric polar output than that provided by the polar control retarder **300A**. In alternative embodiments, the profile **322B** may be omitted and a uniform voltage applied to the further polar control retarder **300B**. Advantageously the reflectivity of the display to off-axis light may be improved when the layer **314** is arranged for privacy mode operation. Security factor, *S* may advantageously be improved.

[0457] FIG. 23C is a schematic side perspective view of a display device **100** providing uniformity for a display user comprising first and second polar control retarders **300A**, **300B** arranged on the output side of the output display polariser **318**. Features of the arrangement of FIG. 23C not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0458] In comparison to the embodiment of FIGS. 23A-B, in the alternative embodiment of FIG. 23A, such an arrangement may be provided for an emissive display. Further, the view angle control element **310** may be provided as a single component, reducing cost and complexity.

[0459] It may be desirable to further improve the security factor of a display.

[0460] FIG. 24A is a schematic side perspective view of a display device **100** providing uniformity for a display user comprising a first polar control retarder **300** with a voltage *V* profile **322** and a second polar control retarder **300** with a profile of alignment layer orientations.

[0461] In the alternative embodiment of FIG. 24A, the polar control retarder **300A** may comprise a twisted liquid crystal layer **314A** as described elsewhere hereinbefore.

[0462] By way of comparison, the further polar control retarder **300B** comprises passive polar control retarder **330**, i.e. at least one passive compensation retarder, and a layer **314B** of liquid crystal material **414B** provided by a switchable liquid crystal retarder **301B**. In general, the polar control retarder **300B** may comprise any configuration of at least one retarder, some examples of

which are present in the devices described below.

[0463] An illustrative arrangement of alignment layers **419BA**, **419BB** for the alternative embodiment of FIG. **24A** will now be further described.

[0464] FIG. **24B** is perspective views of polar control retarder **300B** that may be applied in FIG. **24A**, comprising a homeotropically and homogeneously aligned switchable LC retarder **301** and a negative C-plate passive retarder **330** and wherein the alignment direction of the homeotropic alignment layer **419BA** is common across the lateral direction and wherein the alignment direction of the homogeneous alignment layer **419BB** varies across the lateral direction. Features of the arrangements of FIG. **24B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0465] An exemplary embodiment is illustrated in TABLE 6.

TABLE-US-00006 TABLE 6 Passive polar control retarder 330 Active LC retarder 301 $\Delta n.d/$ Alignment Pretilt/ $\Delta n.d/$ Voltage/ Mode Type nm layers deg nm V Public Negative -900 Homogeneous 2 1000 10.0 Privacy C Homeotropic 88 1.4

[0466] FIG. **24B** is a front view of a first surface alignment layer **419BA** of the display device **100** of FIG. **24A** wherein the angle of in-plane component of the alignment varies along the predetermined direction, for example the x-axis. Features of the arrangement of FIG. **24B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0467] The surface alignment layer **419BA** may have material that provides different alignment orientations **417BLp**, **417BCp**, **417BRp** across the surface alignment layer **419BB**. The operation of the different alignment layer orientations is described further in U.S. Pat. No. 11,079,646 and in U.S. Pat. No. 11,099,448, both of which are herein incorporated by reference in their entireties.

[0468] The operation and function of the arrangement of FIG. **24B** is different to that achieved by the arrangement of FIG. **3A**, other than appropriate phase shifts are provided across the predetermined direction. Advantageously increased uniformity of illumination to on-axis viewers may be increased.

[0469] The alignment orientation **417BLp**, **417BCp**, **417BRp** change monotonically across at least part of the display device **100**. In manufacture, the gradient of the profile may be varied to provide maximum uniformity for different nominal viewing distances of the primary viewer from the display device **100**. For example, a high gradient may be used for a short viewing distance while a lower gradient of profile may be provided for displays arranged to be operated at longer viewing distances. Advantageously uniformity of luminance may be optimised for the passenger **45** and uniformity of security factor may be optimised for the driver **47**.

[0470] FIG. **24C** is a schematic graph illustrating the variation of transmission with polar direction for the polar control retarder of FIG. **24B** in a privacy mode of operation; and FIG. **24D** is a schematic graph illustrating the variation of transmission with polar direction for the polar control retarder of FIG. **24B** in a share mode of operation.

[0471] In comparison with FIG. **4B**, the profile of FIG. **24C** has higher symmetry. Symmetry may be reduced by increasing the average rotation of the alignment layer **417BB**. In comparison to the profile of FIG. **5C**, there is substantially no change with viewing angle and thus colour may be improved.

[0472] Returning to the description of FIG. **24A**, a display may be provided with higher off-axis reduction of luminance levels, increasing polar region of high security factor.

[0473] FIG. **25A** is a schematic side perspective view of a display device **100** providing uniformity for a display user comprising a first polar control retarder **300** with a voltage V profile **322** arranged between the backlight **20** and transmissive SLM **48**, and a reflective polariser, further polar control retarder **300** and further additional polariser arranged to receive light from the SLM **48**. Features of the arrangement of FIG. **25A** not discussed in further detail may be assumed to

correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0474] In comparison to the arrangement of FIGS. 23B-C, increased symmetry of the second polar control retarder **300B** may be achieved, and a larger region with high reflectivity achieved. Advantageously the polar region for high security factor may be increased. The second polar control retarder **300B** may have variable alignment across the predetermined direction as illustrated in FIG. 24B or may have a common alignment across the predetermined direction. Advantageously the size of the polar region of increased reflectivity may be increased and security factor may be increased.

[0475] The embodiments of FIGS. 23A-C, FIG. 24A and FIG. 25A illustrate that the polar control retarders **300A**, **300B** are provided with a common predetermined direction. In other embodiments, not shown, the predetermined directions may be different.

[0476] For example, the predetermined directions may be opposite. Returning to the description of FIG. 1B, one of the polar control retarders **300A**, **300B** may be arranged to provide privacy to the left side of the display device **100** for user **47**, and the other of the polar control retarders **300A**, **300B** may be arranged to provide privacy to the right side of the display device **100** for user **49**.

[0477] In other embodiments, the predetermined directions may be crossed and one of the polar control retarders **300A**, **300B** may be arranged to provide privacy to the left side of the display device **100** for user **47**, and the other of the polar control retarders **300A**, **300B** may be arranged to provide privacy for the vertical direction, for example to minimise windscreen reflections in an automotive vehicle **650**.

[0478] FIG. 25B is a schematic graph illustrating the variation of reflectivity with polar direction for the polar control retarder **300B** and reflective polariser **302** of FIG. 25A and TABLE 6 in a share mode of operation; and FIG. 25C is a graph illustrating a simulated polar profile of the security level, S of the arrangement of FIG. 25 for an ambient illuminance measured in lux that is the same as the head-on display luminance measured in nits and for the polar profile of backlight luminance of FIG. 16A, polar profile of polar control retarder **300B** transmission of FIG. 24C and reflectivity of polar control retarder **300B** and reflective polariser **302** of FIG. 25B.

[0479] In comparison to the arrangement of FIG. 25C, the polar region for desirable security factor is substantially increased. Further the pupillation of the polar control retarders **300A**, **300B** and backlight **20** may achieve advantageously high image security uniformity around the direction **448** across the area of the display device **100**.

[0480] The structure and operation of various alternative backlights **20** that provide desirable illumination characteristics for the switchable privacy display device **100** of the present embodiments will now be described further.

[0481] FIG. 26A is a schematic diagram illustrating a side view of the switchable backlight **20** of FIG. 2 comprising waveguides **1A**, **1B**, a rear reflector **3** and an optical turning film component **50** and outputting light beams **445**, **447** with the angular distributions as illustrated in FIGS. 4A-C; FIG. 26B is a schematic diagram illustrating a front perspective view of an optical turning film component **50** for the backlight **20** of FIG. 26A; and FIG. 26C is a schematic diagram illustrating a side view of an optical turning film component **50**. Features of the embodiments of FIGS. 26A-C not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0482] The backlight **20** comprises: at least one first light source **15A** arranged to provide input light; at least one second light source **15B** arranged to provide input light in an opposite direction from the at least one first light source **15A**; a waveguide arrangement **11** comprising at least one waveguide **1**, the waveguide arrangement **11** being arranged to receive the input light from the at least one first light source and the at least one second light source and to cause light from the at least one first light source and the at least one second light source to exit from the waveguide arrangement **11** by breaking total internal reflection; and an optical turning film component **50**

comprising: an input surface **56** arranged to receive the light exiting from a waveguide **1** through a light guiding surface **8** of the waveguide **1** by breaking total internal reflection, the input surface **56** extending across the plane; and an output surface **58** facing the input surface **56**, wherein the input surface **56** comprises an array of prismatic elements **51**. The prismatic elements **51** may be elongate.

[0483] The waveguide arrangement **11** comprises: a first waveguide **1A** extending across a plane and comprising first and second opposed light guiding surfaces arranged to guide light along the waveguide, the second light guiding surface being arranged to guide light by total internal reflection; and a first input end **2A** arranged between the first and second light guiding surfaces **6A**, **8A** and extending in a lateral direction between the first and second light guiding surfaces **6A**, **8A**; wherein the at least one first light source **15A** is arranged to input light **445** into the first waveguide **1A** through the first input end, and the first waveguide **1A** is arranged to cause light from the at least one first light source **15A** to exit from the first waveguide **1A** through one of the first and second light guiding surfaces **6A**, **8A** by breaking total internal reflection; a second waveguide **1B** extending across the plane arranged in series with the first waveguide **1A** and comprising first and second opposed light guiding surfaces **6B**, **8B** arranged to guide light along the waveguide **1B**, the second light guiding surface **8B** being arranged to guide light by total internal reflection, and a second input end **2B** arranged between the first and second light guiding surfaces **6B**, **8B** and extending in a lateral direction between the first and second light guiding surfaces **6B**, **8B**; wherein the at least one second light source **15B** is arranged to input light **447** into the second waveguide **1B** through the second input end **2B**, and the second waveguide **1B** is arranged to cause light from the at least one second light source **15B** to exit from the second waveguide **1B** through one of the first and second light guiding surfaces **6B**, **8B** by breaking total internal reflection, and wherein the first and second waveguides **1A**, **1B** are oriented so that at least one first light source **15A** and at least one second light source **15B** input light **445**, **447** into the first and second waveguides **1A**, **1B** in opposite directions.

[0484] The optical turning film component **50** comprises: an input surface **56** arranged to receive the light **444A**, **444B** exiting from the waveguide arrangement **11** through a light guiding surface of the at least one waveguide **1A**, **1B** of the waveguide arrangement by breaking total internal reflection, the input surface **56** extending across the plane; and an output surface **58** facing the input surface, wherein the input surface **56** comprises an array of prismatic elements **52**. The prismatic elements each comprise a pair of elongate facets **52** defining a ridge **54** therebetween. Angles $\phi_{\text{sub.A}}$, $\phi_{\text{sub.B}}$ of prism surfaces **53A**, **53B** are provided to direct the nominal light output from waveguides **1A**, **1B** to directions **445**, **447** by refraction and reflection at surfaces **53A**, **53B**. Advantageously desirable illumination directions such as illustrated in FIGS. **4A-F** may be achieved by selection of angles $\phi_{\text{sub.A}}$, $\phi_{\text{sub.B}}$.

[0485] The backlight **20** of FIG. **26A** may provide the exemplary luminance profiles of FIGS. **16A-B** hereinabove. In operation, the light **444A** from the first light source **15A** exits the backlight **20** with a first angular distribution **445** as illustrated in FIG. **16A** and the light from the second light source **15B** exits the backlight **20** with a second angular distribution **457** as illustrated in FIG. **16B** different from the first angular distribution **455**. The first angular distribution **455** may be symmetrical about an axis **199** of symmetry of the backlight **20** and the second angular distribution **457** is asymmetrical about the same axis **199** of symmetry of the backlight **20**. In a left-hand drive vehicle, the asymmetrical distribution **457** may be to the left of the axis **199** of symmetry of the backlight **20** and in a right-hand drive vehicle the asymmetrical distribution **457** may be to right of the axis **199** of symmetry of the backlight **20**.

[0486] Waveguides **1A**, **1B** comprise surface relief features that are arranged to leak some of the guiding light either towards the rear reflector **3** or towards the light turning component **50**. Each waveguide **1A**, **1B** comprise a surface relief **30** arranged on the first side **6A**, **6B** that may comprise prism surfaces **32**, **33**. Further the second sides **8A**, **8B** may further comprise surface relief **31** that

may comprise elongate features or prism features as illustrated in FIG. 14D hereinbelow. In operation the surface reliefs **30**, **31** provide leakage of light **445**, **447** from the waveguide **1A**, **1B** for light guiding along the waveguide **1A**, **1B**.

[0487] FIG. **27** is a schematic diagram illustrating a side view of a switchable backlight **20** comprising a waveguide **1**, first and second light sources **15** at respective opposite input sides of the waveguide **1**, a rear reflector and an optical turning film and outputting light beams for passenger **45** and driver **47** nominal directions. Features of the embodiment of FIG. **27** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0488] The waveguide arrangement **11** comprises: a waveguide **1** extending across a plane, wherein the waveguide **1** is an optical waveguide, and comprising: first and second opposed light guiding surfaces **6**, **8** arranged to guide light along the waveguide **1**, the second light guiding surface being arranged to guide light by total internal reflection **8**, and first and second input ends **2A**, **2B** arranged between the first and second light guiding surfaces **6**, **8** and extending in a lateral direction between the first and second light guiding surfaces **6**, **8**; wherein the at least one first light source **15A** is arranged to input light **445** into the waveguide **1** through the first input end **2A** and the at least one second light source **15B** is arranged to input light **447** into the waveguide **1** through the second input end **2B**, and the waveguide **1** is arranged to cause light from the at least one first light source **15A** and the at least one second light source **15B** to exit from the waveguide **1** through one of the first and second light guiding surfaces **6**, **8** by breaking total internal reflection.

[0489] It may be desirable to pupillate the output of the backlight **20** in a manner similar to the pupillation of the transmission of the polar control retarder **300**.

[0490] FIG. **28A** is a schematic diagram illustrating in side-view operation of a backlight **20** comprising a pupillating light turning film component **50**; and FIG. **28B** is a schematic diagram illustrating in front perspective view, operation of a pupillating light turning film component **50**. Features of the embodiments of FIG. **28A** and FIG. **28B** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0491] In the alternative embodiment of FIG. **28A**, the tilt of the facets **53A**, **53B** are adjusted so that the highest luminance output from the backlight **20** from each part of the light turning component **50** is directed towards a common point **825**.

[0492] In operation, FIG. **28A** illustrates that light rays **415G** are output from the waveguide **1** and directed onto the facets **53B**, **53A** of the light turning component **50**. The inclination angles of the facets **53A**, **53B** are adjusted so that light rays **415U**, **415C**, **415D** are directed towards a common point **825**, wherein the light rays **415** have the maximum luminous intensity of the light cone **425**.

[0493] FIG. **28B** illustrates that the light turning film component may further have curved elongate facets with facet tangents $\gamma_{\text{sub.L}}$, $\gamma_{\text{sub.C}}$, $\gamma_{\text{sub.R}}$ that vary along the predetermined direction. In this manner, pupillation may be provided in both x and y directions from the convergence of light rays **415DR**, **415CR**, **415UR**, **415DC**, **415CC**, **415UC**, **415DL**, **415CL**, **415UL**. Advantageously increased uniformity may be provided in both lateral and vertical directions to the display user **45**.

[0494] The point **825** may be arranged to be at or near the nominal viewing location of the passenger **45**. Advantageously uniformity of the backlight **20** is improved for the user **45**. Such improved uniformity may cooperate with the improved uniformity from the polar control retarder **300** described elsewhere herein.

[0495] Alternative arrangements of backlights **20** suitable for use in the display of FIG. **2A** will now be described.

[0496] FIG. **29** is a schematic diagram illustrating a side view of a switchable backlight **20** comprising a first waveguide **1A**, light turning component **50**, a micro-louvre film **66** and second waveguide **1B**. Features of the embodiment of FIG. **29** not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above,

including any potential variations in the features.

[0497] In comparison to the embodiment of FIG. 26A, the backlight **20** comprises waveguide **1A** arranged between rear reflector **3** and light turning component **50**, the operation of which is described hereinabove.

[0498] In a privacy mode of operation, the light sources **15A** are illuminated and the waveguide **1A** and optical turning film component **50** and provides light rays **445** with cone angle **461**. Such cone angle **461** may be larger than desirable for viewing locations of driver **47** at angles near to angle α of FIG. 1A such that undesirable security factor may be achieved.

[0499] Reflective recirculation polariser **64** may optionally be provided to achieve light recirculation in the backlight **20**. Advantageously efficiency may be increased. The reflective recirculation polariser **64** is different to the reflective polariser **302** of FIG. 25A. Reflective recirculation polariser **208** provides reflection of polarised light from the backlight **20** that has a polarisation that is orthogonal to the electric vector transmission direction of the dichroic input polariser **210** or additional polariser **318A**. Reflective recirculation polariser **64** does not reflect ambient light **604** to a snooper.

[0500] Micro-louvre film **66** (such as ALCF™ from 3M Corporation) may be provided that for input light cone **461** outputs light cone **467** that has a smaller angular size for a given proportional luminance roll-off. Such light cone **467** is substantially transmitted by the waveguide **1B**. Features **30B**, **32B** arranged on the light guiding surfaces **6B**, **8B** of the waveguide **1B** may be arranged to minimise scatter in the predetermined direction.

[0501] In driver **47** illumination and in share mode the light sources **15B** are operated and light cones **457** around light rays **447** are output from the waveguide **1B**. As the illumination is predominately off-axis then a second light turning film component to redirect light rays **447** is not provided.

[0502] Advantageously a switchable backlight **20** with a narrow cone angle to user **45** may be provided. Stray light to driver **47** may be reduced in comparison to the arrangement of FIG. 26A, advantageously achieving improved security factor.

[0503] FIG. 30 is a schematic diagram illustrating a side view of a switchable backlight **20** comprising a first waveguide **1A**, prismatic recirculation films **60**, **62**, a micro-louvre sheet **66** and second waveguide **1B**. Features of the embodiment of FIG. 30 not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0504] In the alternative embodiment of FIG. 30, the light turning film component **50** is omitted and at least one brightness enhancement film **60**, **62** provides recirculation of on-axis scattered light from the rear reflector **3**. In comparison to the alternative embodiment of FIG. 29, the output cone **461** may have a larger size and so losses may increase and off-axis luminance may also increase in the region around the angle α of FIG. 1A. Advantageously cost and complexity may be reduced and uniformity may be increased.

[0505] It may be desirable to provide increased dynamic range and reduced power consumption.

[0506] FIG. 31 is a schematic diagram illustrating a side view of a switchable backlight **20** comprising a mini-LED array **15Aa-n**, prismatic recirculation films **60**, **62**, a micro-louvre sheet **66**, waveguide **1B** and light source **15B**. Features of the embodiment of FIG. 31 not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0507] In comparison to FIG. 30, the embodiment of FIG. 31 comprises array **15Aa-n** of mini-LEDs that are distributed across the area of the backlight **20**. The array **15Aa-n** may provide blue output light and additional diffusers **70** and wavelength conversion sheet **72** may be provided to achieve white light output.

[0508] In operation for illumination of passenger **45**, the array **15Aa-n** may be modulated with image data and aligned to pixels **220**, **222**, **224** of the SLM **48**. Advantageously image contrast may

be increased.

[0509] FIG. 32A is a schematic diagram illustrating a perspective side view of a steerable backlight 20 comprising a waveguide arrangement 11 comprising a stepped waveguide 1, and addressable light source array 15a-n; and FIG. 32B is a schematic diagram illustrating a side view of a display steerable backlight 20 comprising a stepped waveguide, and addressable light source array and rear reflector. Features of the embodiment of FIGS. 32A-B not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0510] Such a stepped waveguide is further described in U.S. Pat. No. 9,519,153 and in U.S. Pat. No. 10,054,732, both of which are herein incorporated by reference in their entireties.

[0511] Waveguide 1 comprises an input end 2, a reflective end 4 and first and second light guiding surfaces 6, 8 arranged between the input end 2 and reflective end 4. The second light guiding surface 8 may be a planar surface and the first light guiding surface 6 may comprise a stepped structure comprising steps 12 and intermediate regions 10 that may be planar.

[0512] In operation, light from at least some of light sources 15a-n is input at the input end 2 and guided substantially without loss to the reflective end 4. Reflected light rays are guided back towards the steps 12 by means of the surface 8, 10 at which point they are extracted from the waveguide through total internal reflection or by refraction. Considering FIG. 32A, light sources 15 in region 31 are directed towards driver 47 and light sources in region 33 are directed towards the passenger 45.

[0513] Considering FIG. 32B, refracted light rays 445b are incident on rear reflector 3 comprising reflective facets 34, 36 and directed towards passenger 45, similarly refracted light rays 447b (not shown) are directed towards driver 47. Advantageously output brightness is increased.

[0514] The light extraction features 12 may be curved and optical pupils 325, 327 may be provided towards passenger 45, and driver 47. Advantageously image uniformity may be increased.

[0515] FIG. 32C is a schematic diagram illustrating a perspective rear view of a steerable backlight 20 comprising a stepped waveguide 1, addressable light source array 15 spatial light modulator 48 and polar control retarder 300 to illuminate a driver 47 and passenger 45. Features of the embodiment of FIG. 32C not discussed in further detail may be assumed to correspond to the features with equivalent reference numerals as discussed above, including any potential variations in the features.

[0516] In operation for a moving passenger 47 with movement of measured distance 545 from a reference location, the light sources 15a-n are controlled such that region 33 is adjusted to move location by distance 542 in correspondence to the distance of movement 540. The light sources 15 are controlled accordingly such that the input illuminated region 31 is moved by distance 542. Optical window 325 is then maintained at a location near to the passenger, and light leakage to window 327 is minimised. As described elsewhere herein, the polar control retarder 300 may similarly be controlled to achieve desirable luminance reduction at the driver 47 measured distance of movement 540 from a reference location. Advantageously luminance to passenger 45 is increased and security factor to driver 47 is increased.

[0517] In share mode operation, all of the light sources 15a-n may be operated to achieve wide angle operation, together with share mode operation of the polar control retarder 300, described elsewhere herein.

[0518] A thin and steerable backlight for a switchable privacy display with high image security may advantageously be achieved.

[0519] While various embodiments in accordance with the principles disclosed herein have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with any claims and their equivalents issuing from this disclosure. Furthermore, the above advantages and

features are provided in described embodiments, but shall not limit the application of such issued claims to processes and structures accomplishing any or all of the above advantages.

[0520] Additionally, the section headings herein are provided for consistency with the suggestions under 37 CFR 1.77 or otherwise to provide organizational cues. These headings shall not limit or characterize the embodiment(s) set out in any claims that may issue from this disclosure. Specifically and by way of example, although the headings refer to a “Technical Field,” the claims should not be limited by the language chosen under this heading to describe the so-called field. Further, a description of a technology in the “Background” is not to be construed as an admission that certain technology is prior art to any embodiment(s) in this disclosure. Neither is the “Summary” to be considered as a characterization of the embodiment(s) set forth in issued claims. Furthermore, any reference in this disclosure to “invention” in the singular should not be used to argue that there is only a single point of novelty in this disclosure. Multiple embodiments may be set forth according to the limitations of the multiple claims issuing from this disclosure, and such claims accordingly define the embodiment(s), and their equivalents, that are protected thereby. In all instances, the scope of such claims shall be considered on their own merits in light of this disclosure, but should not be constrained by the headings set forth herein.

Claims

1-40. (canceled)

41. A display apparatus comprising: a display device comprising: a spatial light modulator arranged to output spatially modulated light; a display polariser arranged on a side of the spatial light modulator, the display polariser being a linear polariser; a view angle control arrangement comprising: an additional polariser arranged on the same side of the spatial light modulator as the display polariser, the additional polariser being a linear polariser; and at least one polar control retarder arranged between the display polariser and the additional polariser, wherein the at least one polar control retarder comprises a switchable liquid crystal retarder comprising a layer of liquid crystal material; a plurality of surface alignment layers disposed adjacent to the layer of liquid crystal material and on opposite sides thereof, the surface alignment layers each being arranged to provide alignment in the liquid crystal material, the liquid crystal material being provided with a twist; and a first electrode arrangement and a second electrode arrangement disposed on the layer of liquid crystal material and being arranged to provide an electric field to the layer of liquid crystal material; and a control system arranged to provide voltages to the first electrode arrangement and the second electrode arrangement for controlling a direction of minimum light transmission of the view angle control arrangement.

42. The display apparatus of claim 41, further comprising a viewer tracking system arranged to track the location of a viewer, wherein the control system is arranged to vary the voltages to at least one of the first electrode arrangement or the second electrode arrangement based on the tracked location of the viewer.

43. The display apparatus of claim 41, wherein the control system is arranged to accept a user input indicating a direction of minimum light transmission of the view angle control arrangement, and the control system is arranged to vary the voltages to at least one of the first electrode arrangement or the second electrode arrangement based on the user input.

44. The display apparatus of claim 41, wherein the magnitude of the electric field perpendicular to the layer of liquid crystal material changes monotonically along a predetermined axis across at least part of the display device.

45. The display apparatus of claim 44, wherein the magnitude of the electric field changes monotonically along a predetermined axis across at least part of the display device so that directions of minimum light transmission of the view angle control arrangement from points of the at least part of the display device are directed towards a common off-axis point in front of the

display device.

46. The display apparatus of claim 41, wherein the first electrode arrangement comprises a continuous electrode extending over the entirety of the layer of liquid crystal material, the continuous electrode having at least one contact disposed at each of opposite ends of the continuous electrode along the predetermined axis and arranged to supply respective voltages that create a voltage profile on the continuous electrode along the predetermined axis for providing the electric field perpendicular to the layer of liquid crystal material.

47. The display apparatus of claim 46, wherein at least one of the contacts comprises a conductive bar extending perpendicular to the predetermined axis.

48. The display apparatus of claim 44, wherein the first electrode arrangement comprises plural electrodes separated along the predetermined axis, the plural electrodes being arranged to supply different voltages that change monotonically along the predetermined axis.

49. The display apparatus of claim 46, wherein the second electrode arrangement is arranged to supply a common voltage.

50. The display apparatus of claim 49, wherein the second electrode arrangement comprises a continuous electrode extending over the entirety of the layer of liquid crystal material.

51. The display apparatus of claim 41, wherein the magnitude of the electric field is uniform across the entirety of the display device.

52. The display apparatus of claim 51, wherein each of the first electrode arrangement and the second electrode arrangement comprises a continuous electrode extending over the entirety of the layer of liquid crystal material.

53. The display apparatus of claim 41, wherein the surface alignment layers are each arranged to provide homogenous alignment in the adjacent liquid crystal material, the surface alignment layers having an in-plane component in the plane of the layer of liquid crystal material having an angle that changes monotonically along a predetermined axis across at least part of the display device.

54. The display apparatus of claim 41, wherein the twist is in a range from 60° to 120° .

55. The display apparatus of claim 41, wherein the twist is in a range from 70° to 90° .

56. The display apparatus of claim 41, wherein an electric vector transmission direction of the additional polariser is oriented at a non-zero angle with respect to an electric vector transmission direction of the display polariser.

57. The display apparatus of claim 56, wherein the non-zero angle is the same as the twist.

58. A display apparatus comprising: a display device comprising: a spatial light modulator arranged to output spatially modulated light; a display polariser arranged on a side of the spatial light modulator, the display polariser being a linear polariser; a view angle control arrangement comprising: a plurality of surface alignment layers disposed adjacent to a layer of liquid crystal material of a switchable liquid crystal retarder and on opposite sides thereof, the surface alignment layers each being arranged to provide alignment in the liquid crystal material, the liquid crystal material being provided with a twist; and a first electrode arrangement and a second electrode arrangement disposed on the layer of liquid crystal material and being arranged to provide an electric field to the layer of liquid crystal material; and a control system arranged to provide voltages to the first electrode arrangement and the second electrode arrangements for controlling a direction of minimum light transmission of the view angle control arrangement.

59. The display apparatus of claim 58, wherein the twist is at least one of in a range from 60° to 120° or in a range from 70° to 90° .

60. The display apparatus of claim 58, further comprising an additional polariser arranged on the same side of the spatial light modulator as the display polariser, wherein an electric vector transmission direction of the additional polariser is oriented at a non-zero angle with respect to an electric vector transmission direction of the display polariser, and the non-zero angle is the same as the twist.
